

SECTION 1 – SYSTEM CONFIGURATION

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1-1 INTRODUCTION

Magnetic Resonance (MR) system upgrades to Signa LX With EXCITE Technology 1.5T, Signa LX with ACGD, Signa Horizon, and Signa CVMR are documented throughout this publication. Refer to the appropriate section listed below for overview of the upgrade content, refer to the remainder of this publication for upgrade requirements. System cable impacts are provided in SECTION 6 INTERCONNECT DATA.

Note

Systems with Oxford or S-1 magnet must upgrade to the LCC magnet before or concurrent with system upgrade.

- SECTION 1-2 UPGRADES TO 1.5T SIGNA LX WITH EXCITE TECHNOLOGY
 - Signa Horizon LX SmartSpeed with EXCITE Technology for 1.5T
 - Signa Horizon LX HiSpeed Plus with EXCITE Technology for 1.5T
 - Signa Horizon LX EchoSpeed Plus with EXCITE Technology for 1.5T
- SECTION 1-3 UPGRADES TO SIGNA LX WITH ACGD
 - Signa Horizon LX SmartSpeed with ACGD for 1.5T or 1.0T
 - Signa Horizon LX HiSpeed Plus with ACGD for 1.5T or 1.0T
 - Signa Horizon LX EchoSpeed Plus with ACGD for 1.5T
- SECTION 1-4 SIGNA LX SYSTEM OPTIONS
- SECTION 1-5 UPGRADES TO SIGNA LX WITH SGD SYSTEM
- SECTION 1-6 SIGNA HORIZON 5.5 SYSTEM OPTIONS
- Refer to Direction 2246012 Pre-Sale/Site Survey For Upgrades to CXK4 Magnet With WideOpen Enclosure which must be completed and signed-off by Local Customer Team to qualify a site for magnet upgrade if such an upgrade is being considered. Direction 2246012 requires input supplied by the customer and by GE. Electronic copy of Direction 2246012 is available to GEMS personnel via the GE Intranet at <http://3.87.118.28/optec3/common/upgradeinfo/upgrade.htm>.

1-2 UPGRADES TO 1.5T SIGNA LX WITH EXCITE TECHNOLOGY

Illustration 1-17 shows the system's major equipment after a 1.5T Signa LX with EXCITE Technology upgrade is complete. The upgrade content is dependent on the configuration of the system being upgraded and the final Signa LX with EXCITE Technology product configuration. The following sub-sections provide an overview of the various Signa LX with EXCITE Technology upgrade hardware.

Note

At this time upgrades to EXCITE Technology are not available for systems with 1.0T Magnets or systems with CRM Body Coils.

WARNING!

ALL SYSTEMS UPGRADING TO SYSTEM CONFIGURATION WITH THE ACGD/PDU CABINET NEED TO INVESTIGATE THAT THE FACILITY TRANSFORMER AND FEEDER WIRES ARE CORRECTLY SIZED. ALSO THE MAIN DISCONNECT CONTROL NEEDS TO BE INVESTIGATED FOR CORRECTLY SIZED WIRES AND RATED COMPONENTS TO MEET THE UPGRADE POWER REQUIREMENTS. REFER SECTION 5 – POWER REQUIREMENTS.

SYSTEMS UPGRADING TO SYSTEM CONFIGURATION WITH ACGD/PDU CABINET WITH AN UNINTERRUPTIBLE POWER SUPPLY (UPS) PROVIDING POWER TO THE ENTIRE SYSTEM NEED TO MAKE SURE THE UPS OPERATION PARAMETERS ARE COMPATIBLE WITH THE SIGNA SYSTEM WITH ACGD POWER AND REGULATION DEMANDS. REFER SECTION 5 – POWER REQUIREMENTS.

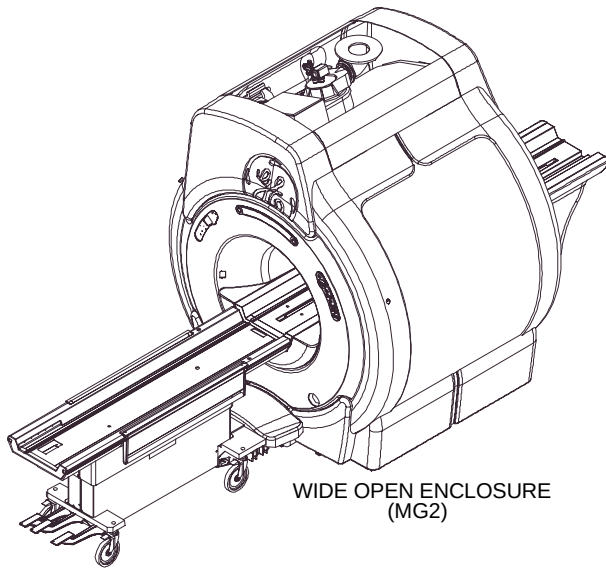
Note

Only catalogs which include hardware are documented in this publication. Software only catalogs are not included, existing system site parameters apply for software only upgrades.

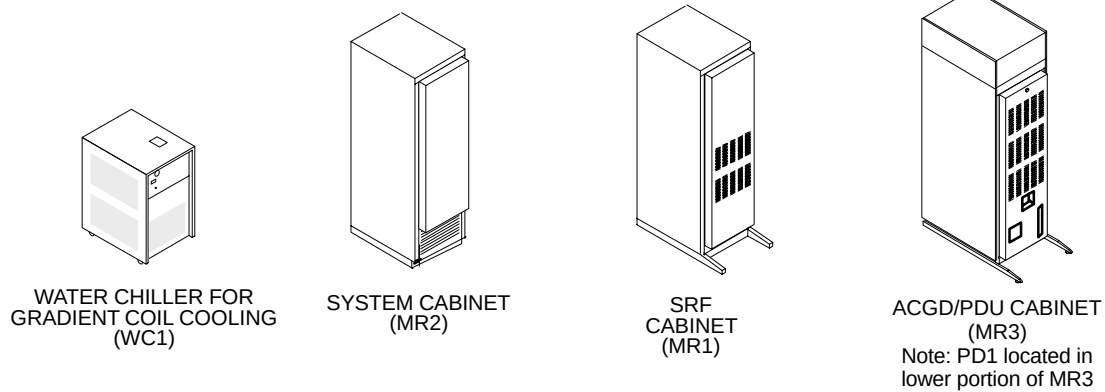
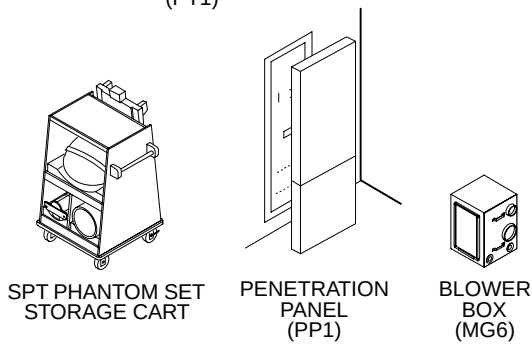
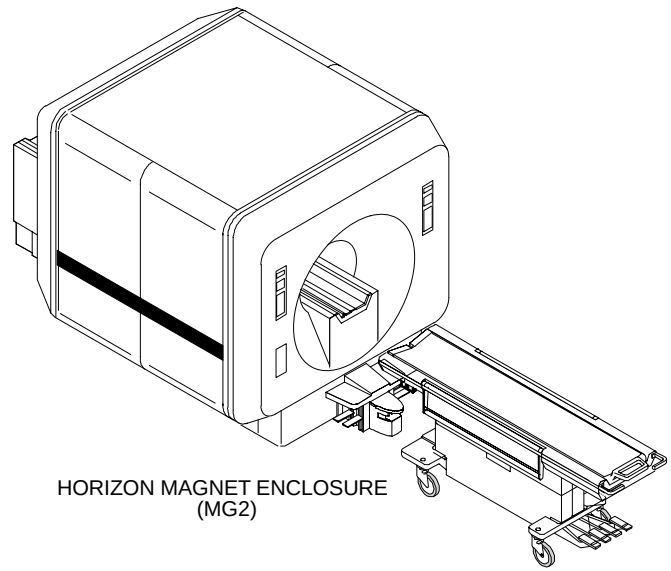
Note

FOR USA, FIXED SITE ONLY CONSIDERING MAGNET UPGRADE: Refer to *Direction 22461012 Pre-Sales/Site Survey For Upgrade To CXK4 Magnet With WideOpen Enclosure* for the many environmental factors and logistics that need to be understood to qualify a site for a magnet upgrade. The Magnet Upgrade Option requires the MR system to be operating at Release 8.X prior or coincident with the Magnet Upgrade option.

1-2 UPGRADES TO 1.5T SIGNA LX WITH EXCITE TECHNOLOGY (Continued)



OR



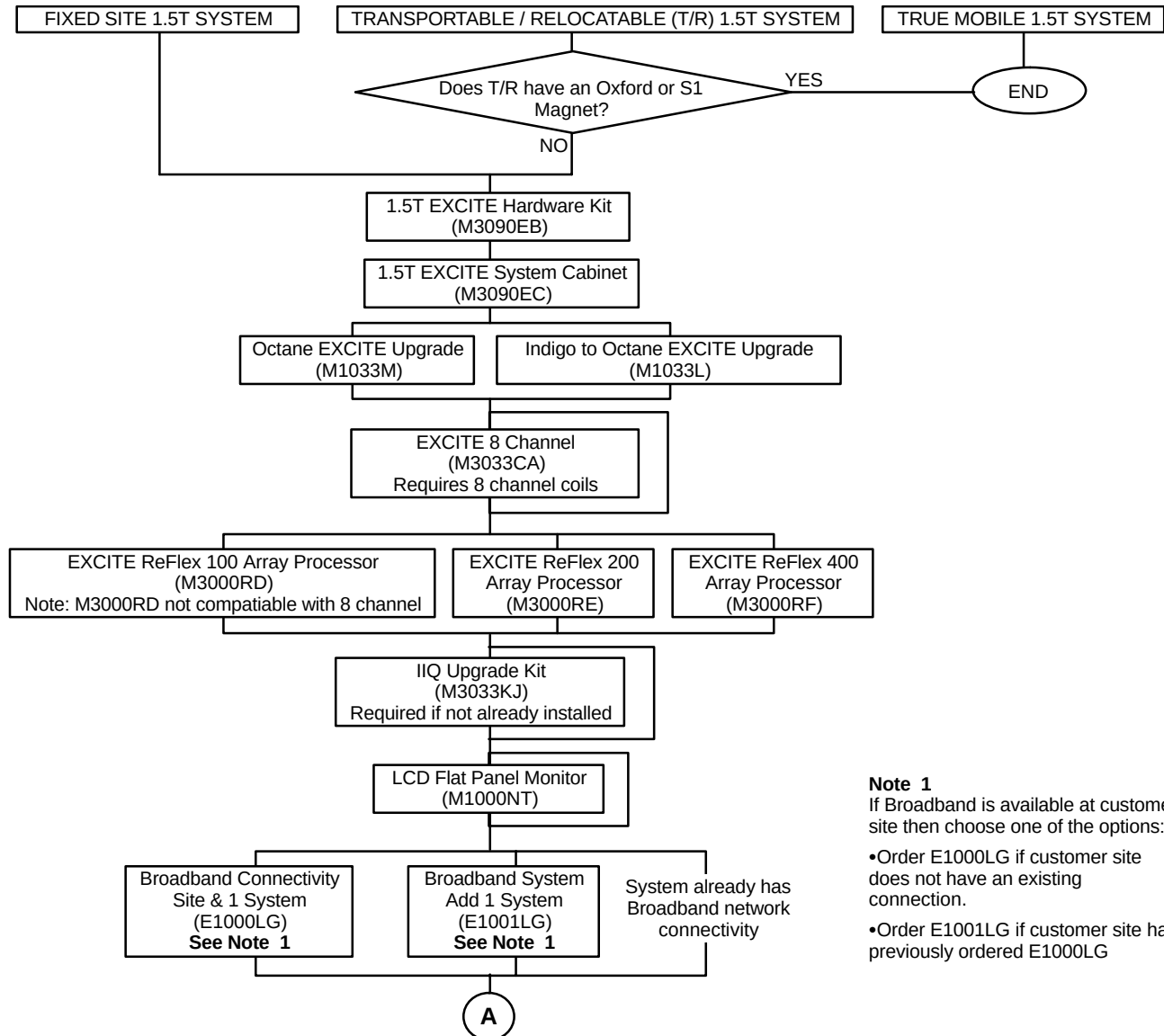
SIGNA LX WITH EXCITE TECHNOLOGY SYSTEM MAJOR EQUIPMENT
ILLUSTRATION 1-1

1-2-1 Signa Horizon LX System

Illustrations 1-2 and 1-3 lists the upgrade catalogs for Signa LX with ACGD or SGD and Signa MR/i system to Signa LX With EXCITE Technology. Illustration 1-4 shows the cabinets replaced and modified by the listed catalogs.

Note

Systems with Oxford or S-1 magnet must upgrade to the LCC magnet before or concurrent with system upgrade.



Note 1

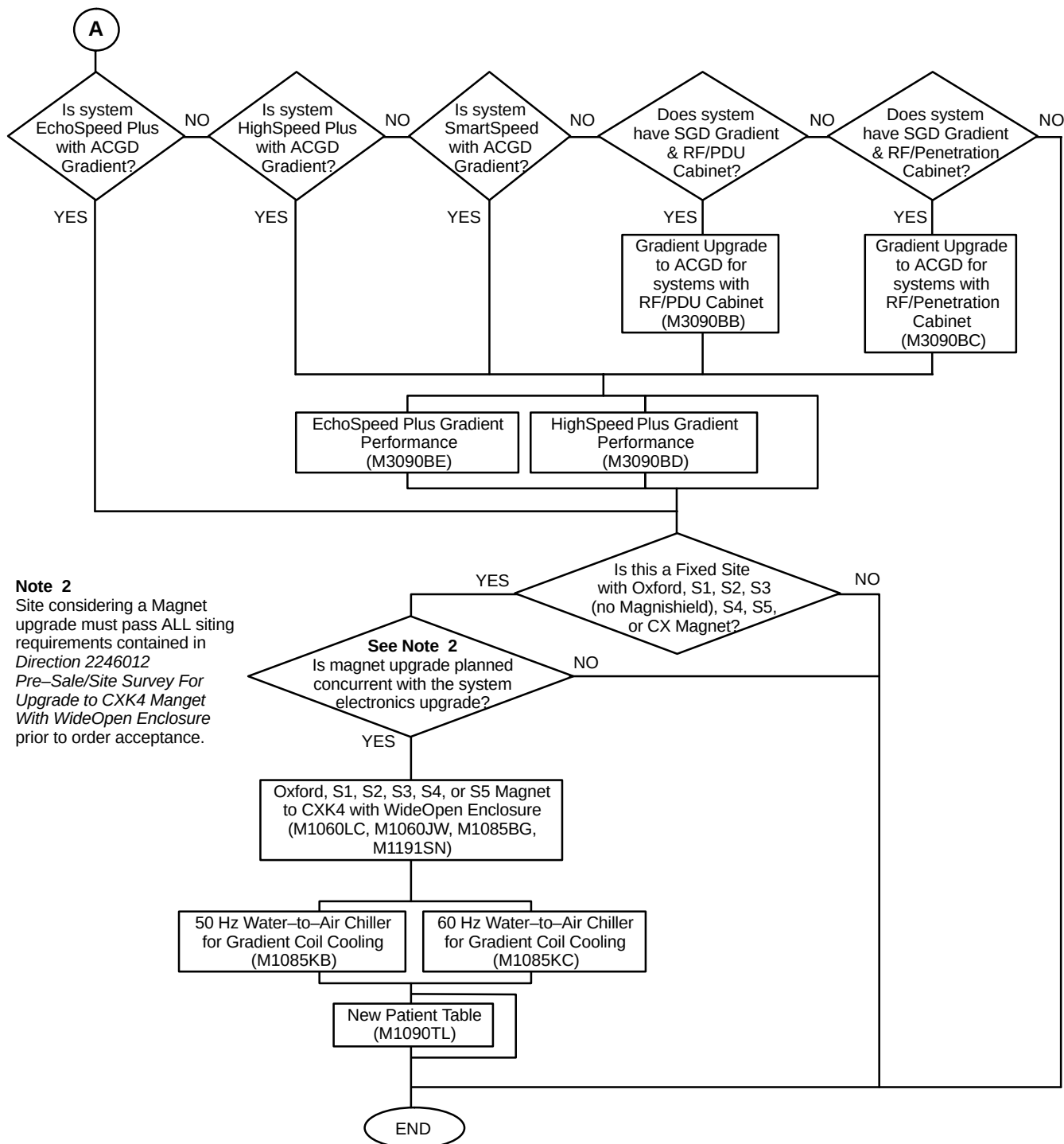
If Broadband is available at customer site then choose one of the options:

- Order E1000LG if customer site does not have an existing connection.
- Order E1001LG if customer site has previously ordered E1000LG

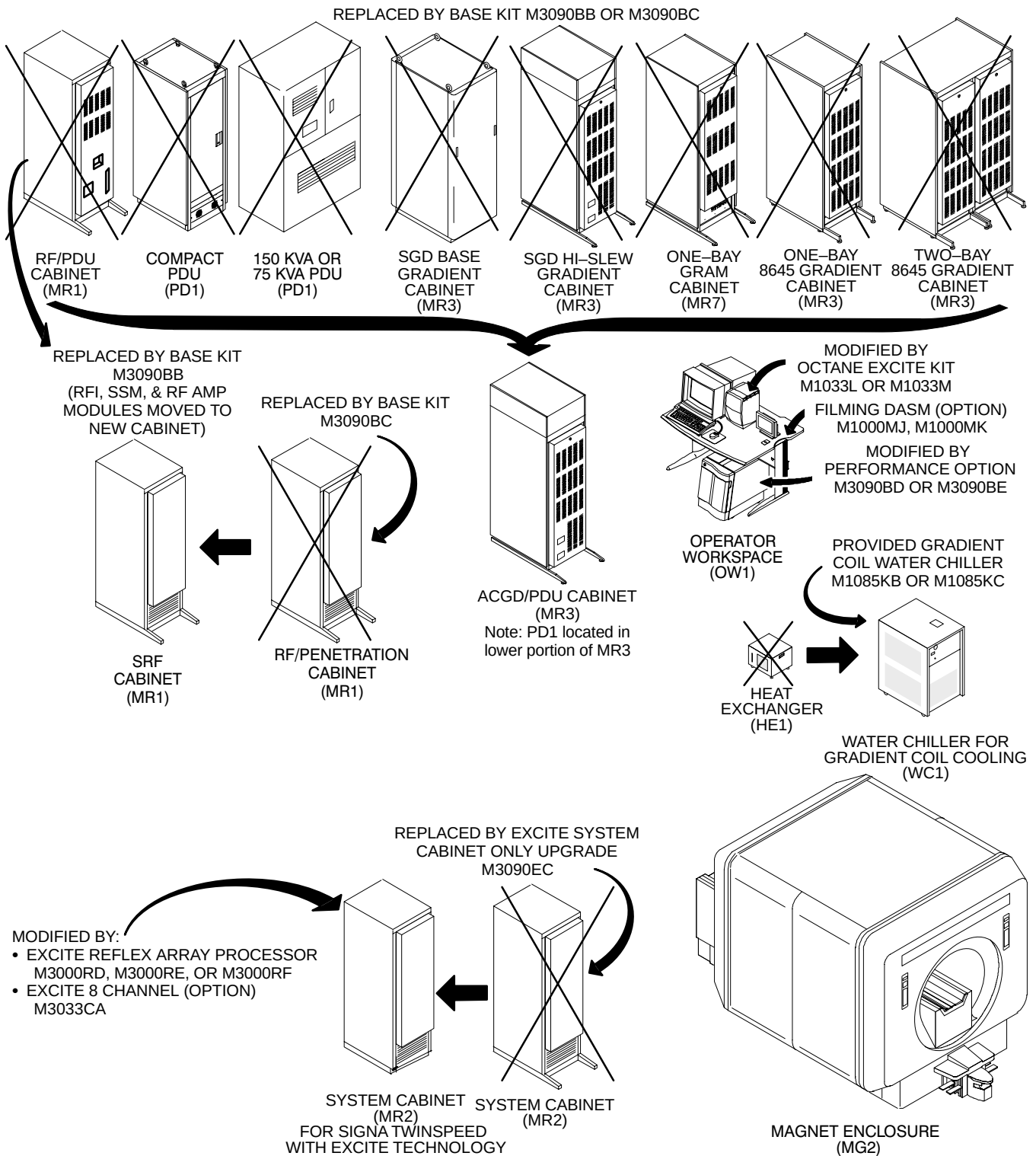
1.5T SIGNA LX TO SIGNA LX WITH EXCITE TECHNOLOGY CATALOGS

ILLUSTRATION 1-2

1-2-1 Signa Horizon LX System(Continued)



1-2-1 Signa Horizon LX System(Continued)



SIGNA HORIZON LX SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

ILLUSTRATION 1-4

1-2-2 Signa Horizon 5.X System

Illustrations 1-5 through 1-7 lists the upgrade catalogs for Signa Horizon 5.X system to Signa Horizon LX With EXCITE Technology. Illustrations 1-8 and 1-9 how the cabinets replaced or modified and new equipment added by the listed catalogs.

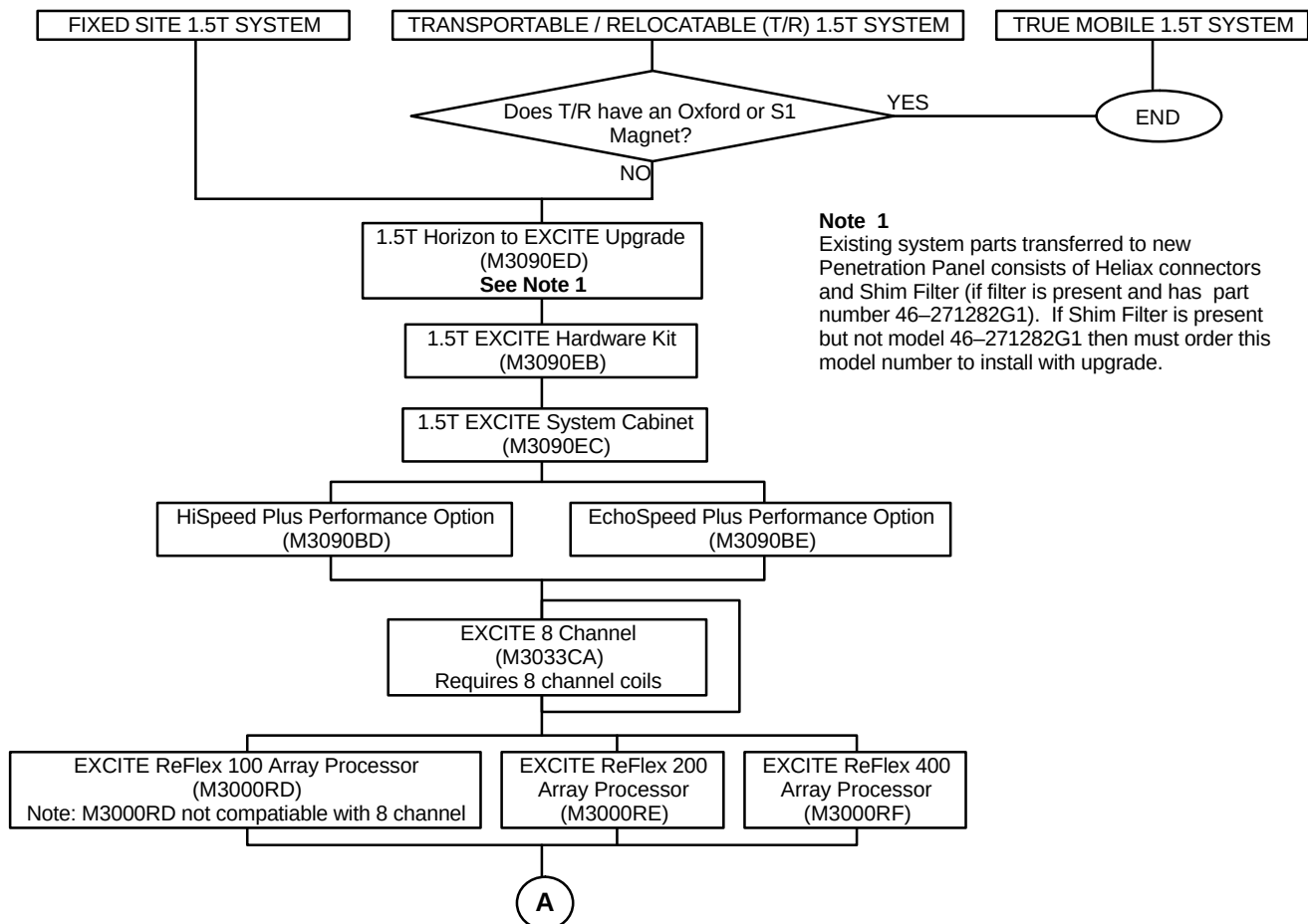
Refer to Section 1-4 SIGNA HORIZON LX SYSTEM OPTIONS for additional option catalogs.

Note

Signa Horizon 5.5 Software only systems have a 55 cm Body Coil and require upgrade catalogs defined in Section 1-2-3, Signa Horizon 5.X Software Only & Signa Advantage 5.X System.

Note

Systems with Oxford or S-1 magnet must upgrade to the LCC magnet before or concurrent with system upgrade.

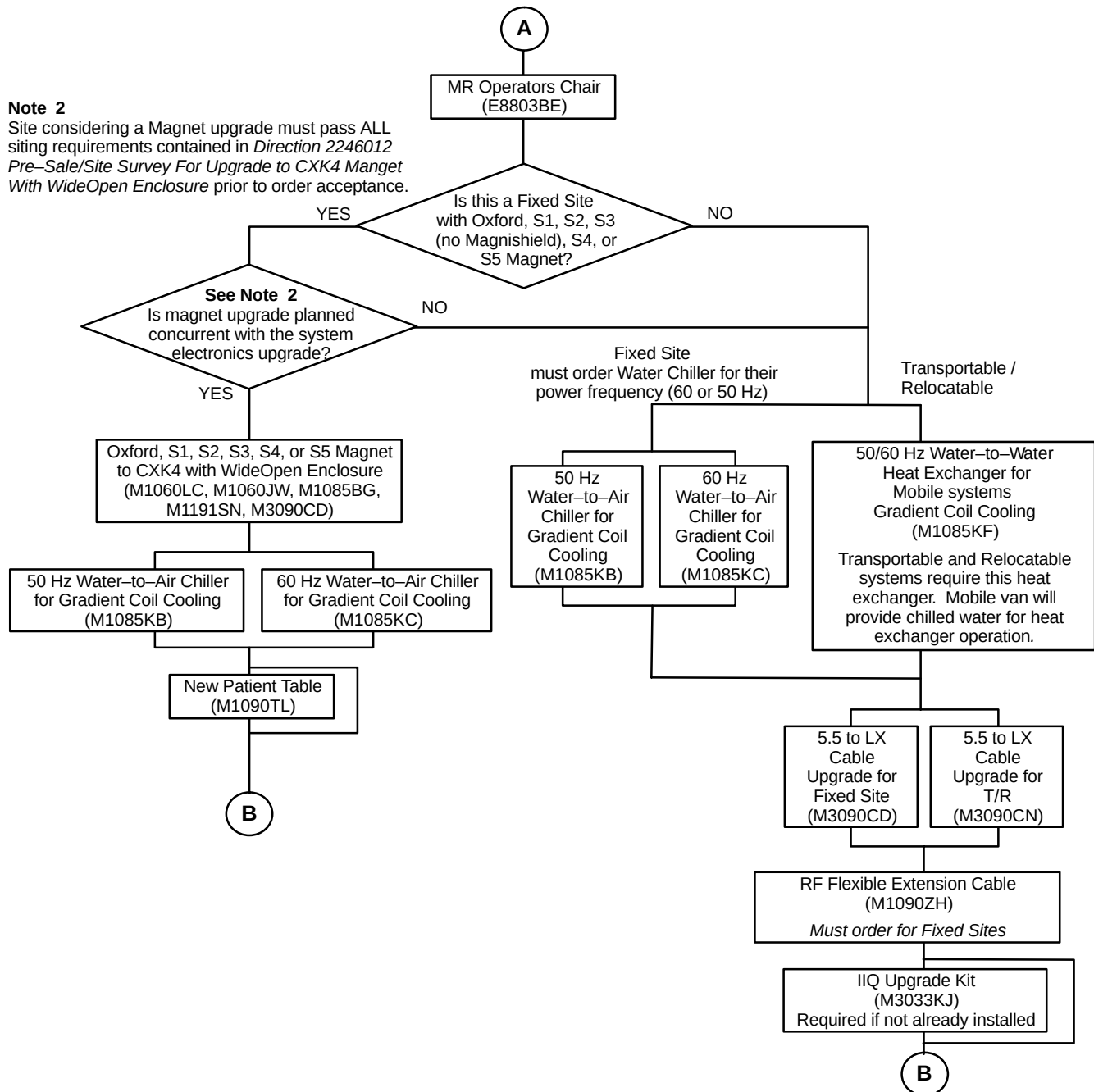


1.5T SIGNA HORIZON 5.X TO SIGNA LX WITH EXCITE TECHNOLOGY CATALOGS
ILLUSTRATION 1-5

1-2-2 Signa Horizon 5.X System (Continued)

Note 2

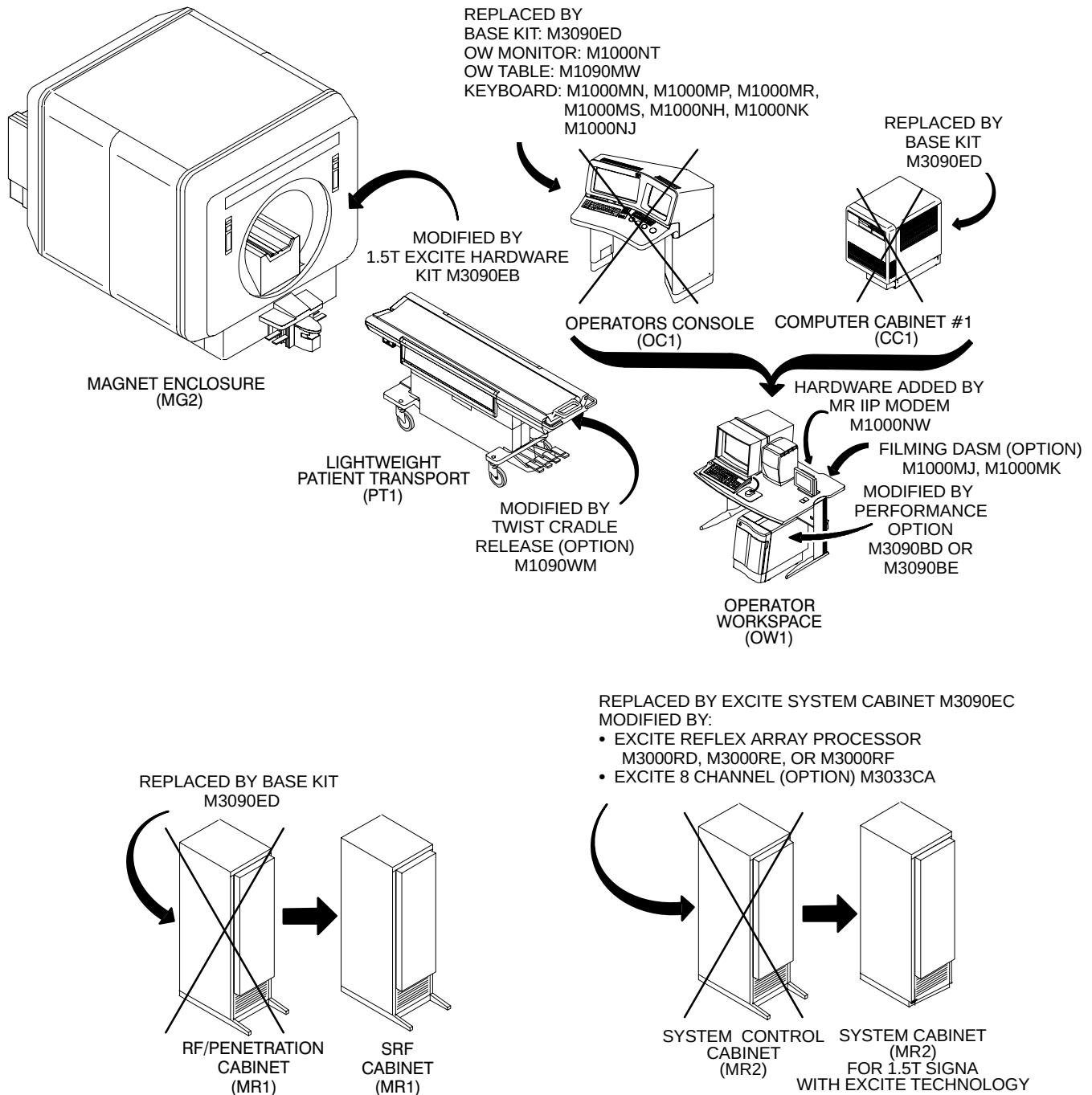
Site considering a Magnet upgrade must pass ALL siting requirements contained in *Direction 2246012 Pre-Sale/Site Survey For Upgrade to CXK4 Magnet With WideOpen Enclosure* prior to order acceptance.



SIGNA HORIZON 5.X TO SIGNA LX WITH EXCITE TECHNOLOGY CATALOGS

ILLUSTRATION 1-6

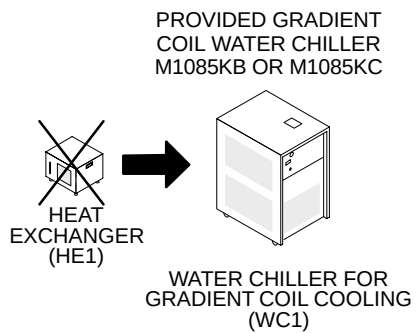
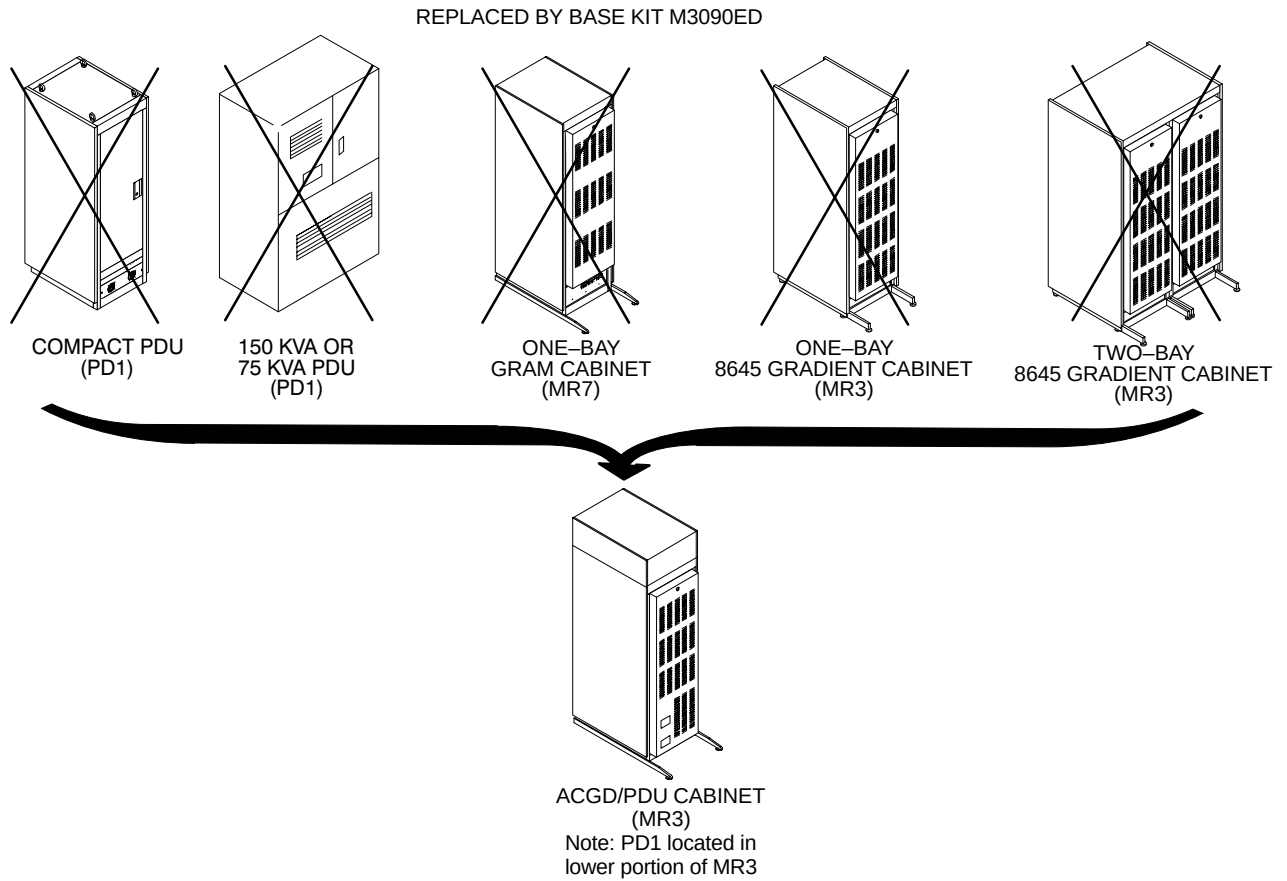
1-2-2 Signa Horizon 5.X System (Continued)



SIGNA HORIZON 5.X SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

ILLUSTRATION 1-8

1-2-2 Signa Horizon 5.X System (Continued)



SIGNA HORIZON 5.X SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

ILLUSTRATION 1-9

1-2-3 Signa Advantage 1.5T 5.X & 4.X

Illustrations 1-10 through 1-12 list the upgrade catalogs for 1.5T Signa Horizon 5.X software only system, 1.5T Signa Advantage 5.X hardware system, and 1.5T Signa 4.X system to 1.5T Signa LX with EXCITE Technology. Signa 5.1-5.4 and 4.X hardware systems are the following configurations:

- Signa Horizon 5.X Software only systems which have a 55 cm Body Coil
- Signa Advantage 5.X systems identified by the Genesis Computer, 1-bay System Cabinet, and 8607 Gradient Cabinet.
- Signa Advantage 4.X hardware systems are Signa Advantage 4.5 to 4.8 which can be identified by RF Cabinet model number 46-306900 and Signa Advantage 4.0 to pre-4.5 which can be identified by RF Cabinet model number 46-282900.

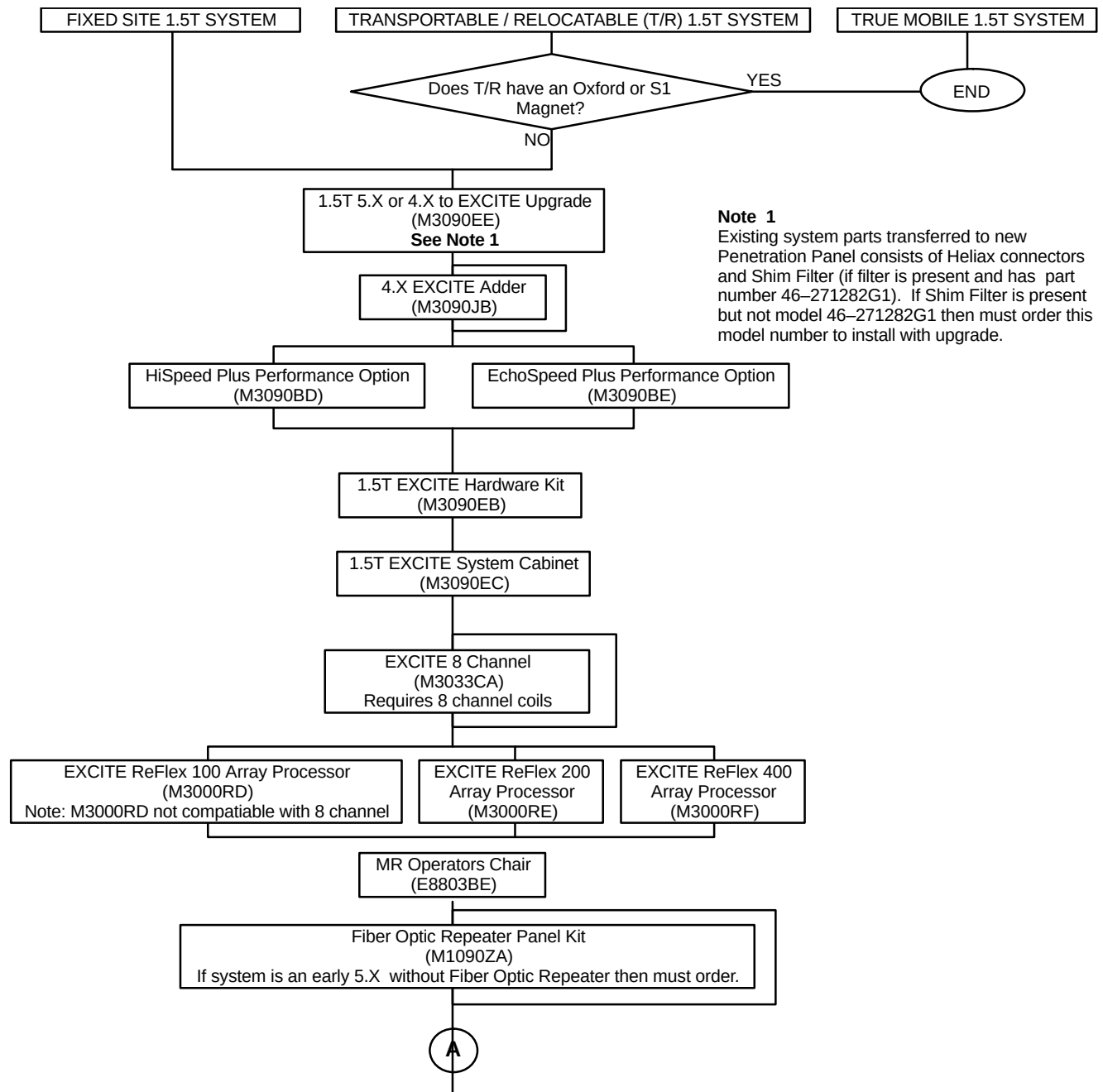
The flowchart shown in Illustrations 1-10 through 1-12 lists the Signa Horizon 5.X Software Only, Advantage 5.X and 4.X to 1.5T Signa LX with EXCITE Technolog upgrade catalogs. The equipment replaced and modified by the catalogs listed in the flowchart are shown in the illustrations as note below:

- Signa Horizon 1.5T 5.X Software Only, Advantage 5.X see Illustrations 1-13 and 1-14
- Signa Horizon 1.5T 4.X see Illustrations 1-15 and 1-16.

Note

Systems with Oxford or S-1 magnet must upgrade to the LCC magnet before or concurrent with system upgrade.

1-2-3 Signa Advantage 1.5T 5.X & 4.X (Continued)

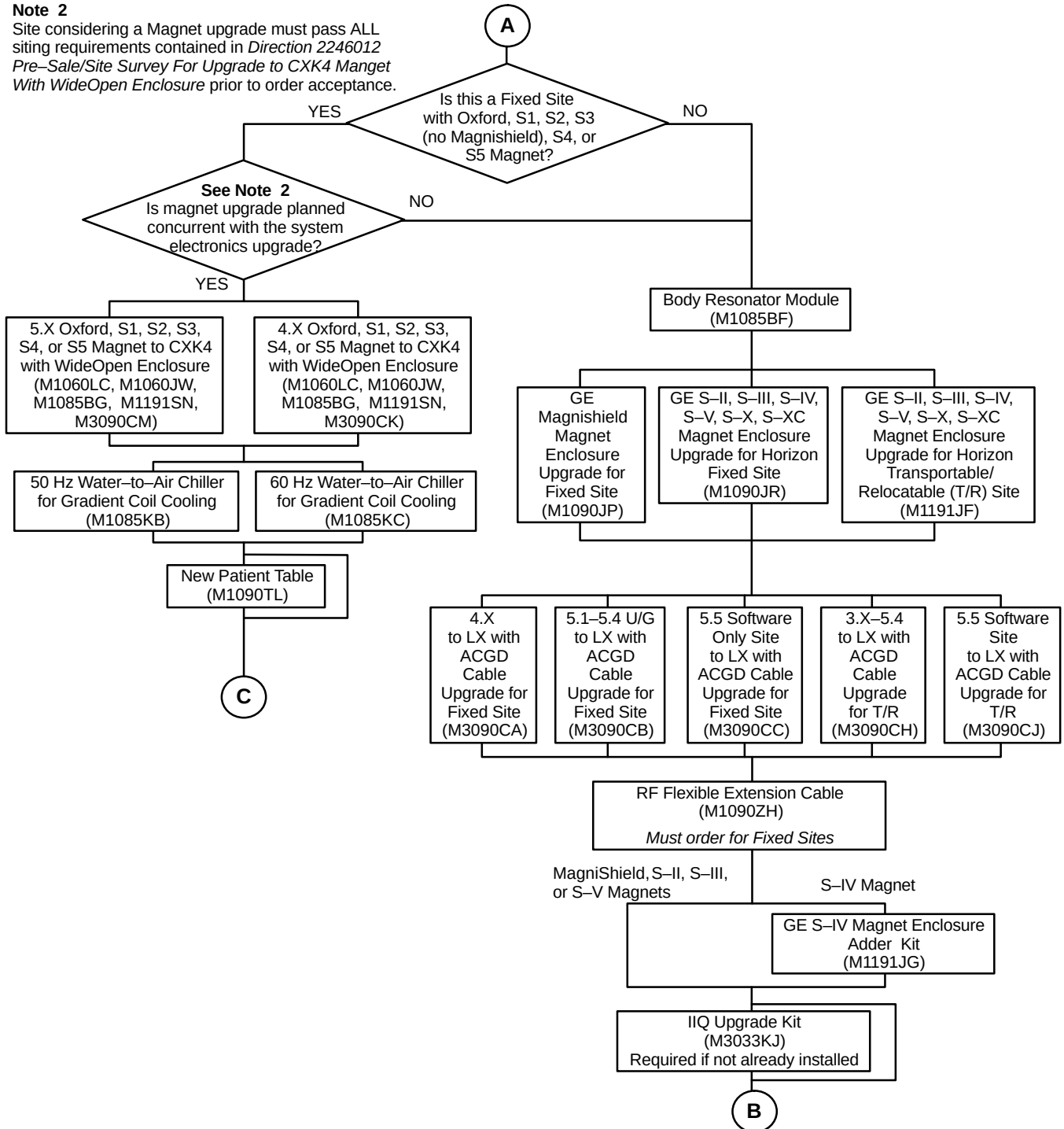


1.5T SIGNA HORIZON 5.X SOFTWARE ONLY & SIGNA ADVANTAGE 5.X TO SIGNA LX WITH EXCITE TECHNOLOGY CATALOGS
ILLUSTRATION 1-10

1-2-3 Signa Advantage 1.5T 5.X & 4.X (Continued)

Note 2

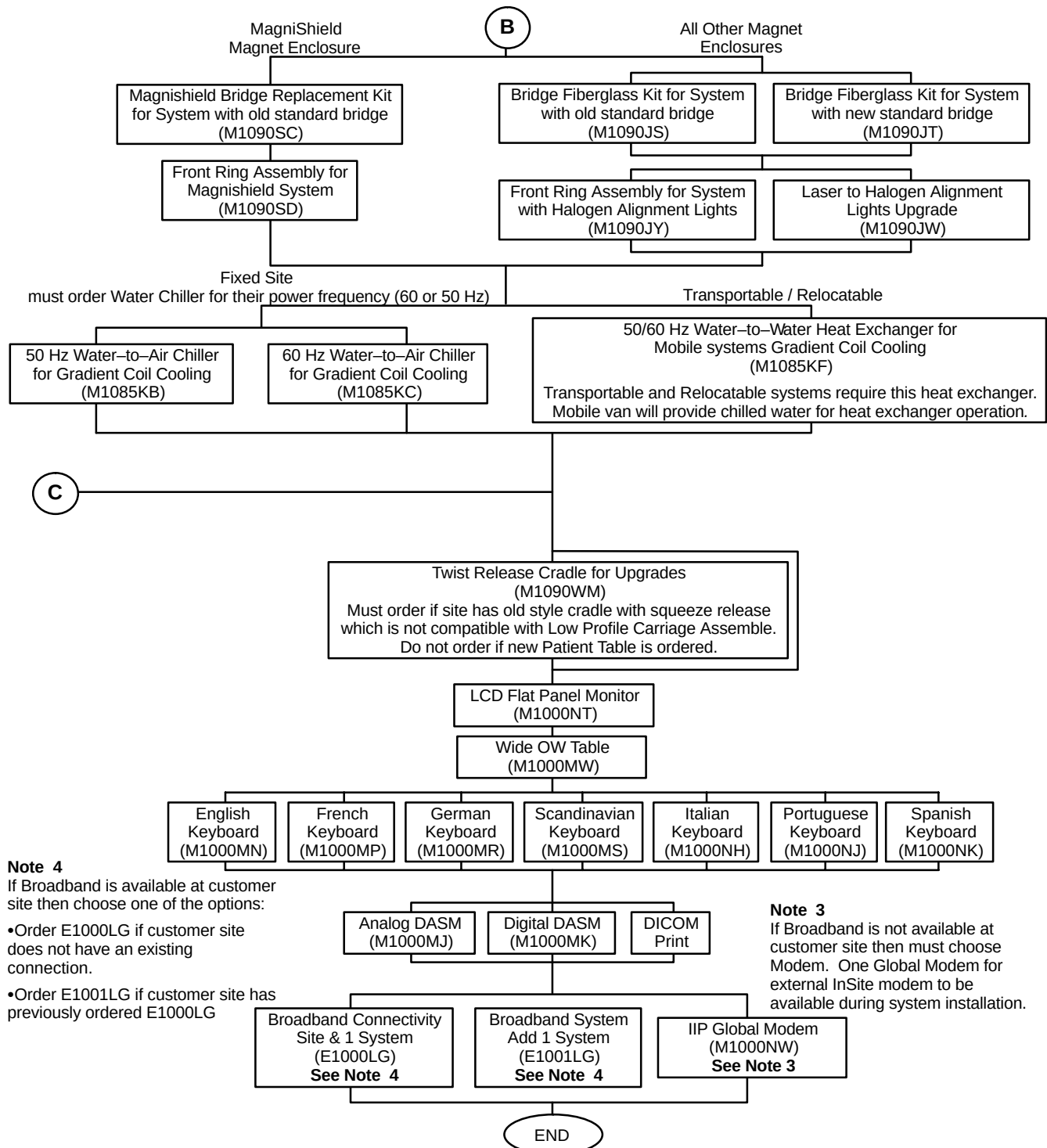
Site considering a Magnet upgrade must pass ALL siting requirements contained in *Direction 2246012 Pre-Sale/Site Survey For Upgrade to CXK4 Magnet With WideOpen Enclosure* prior to order acceptance.



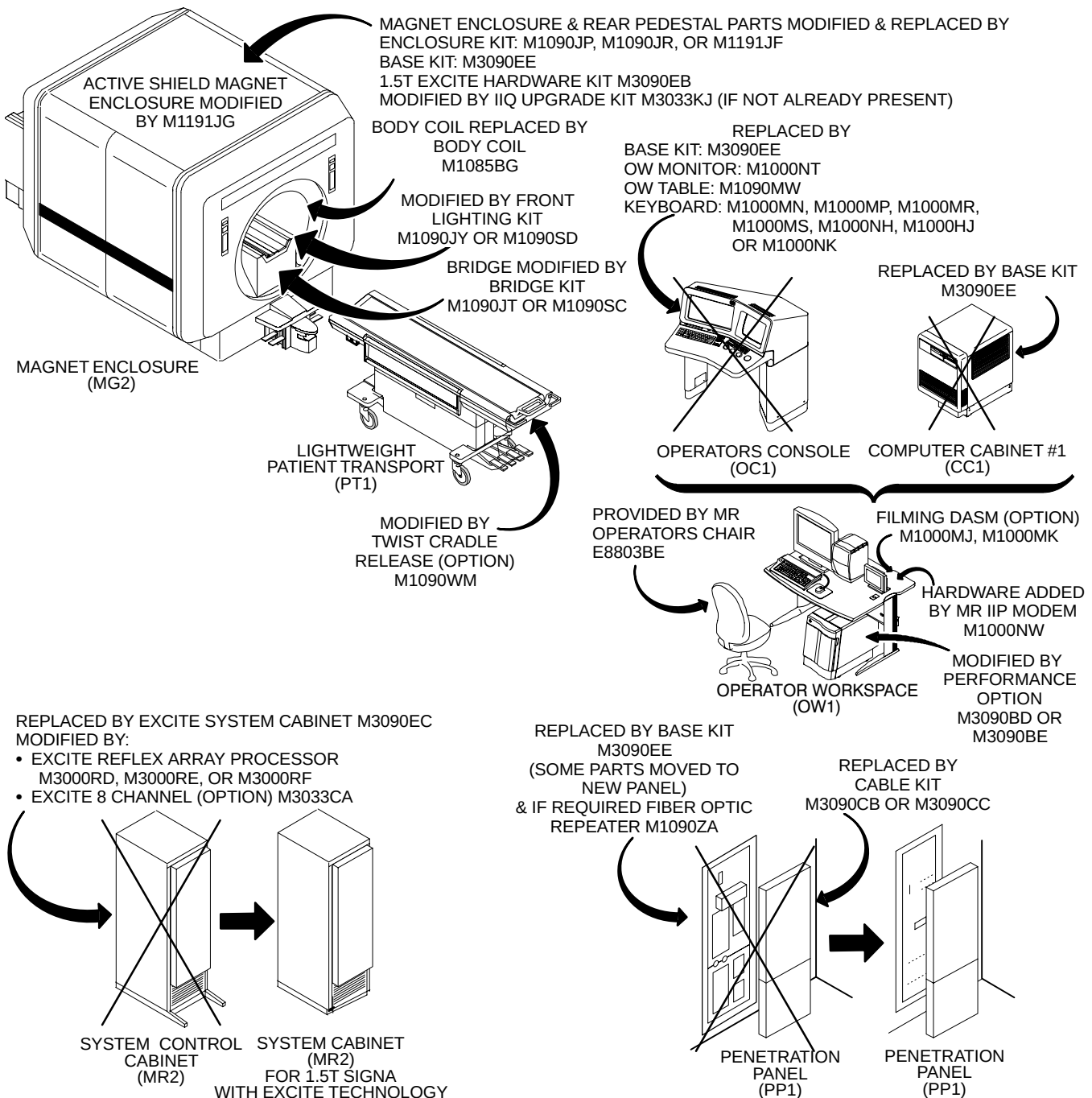
SIGNA HORIZON 5.X SOFTWARE ONLY & SIGNA ADVANTAGE 5.X TO SIGNA LX WITH EXCITE TECHNOLOGY CATALOGS

ILLUSTRATION 1-11

1-2-3 Signa Advantage 1.5T 5.X & 4.X (Continued)

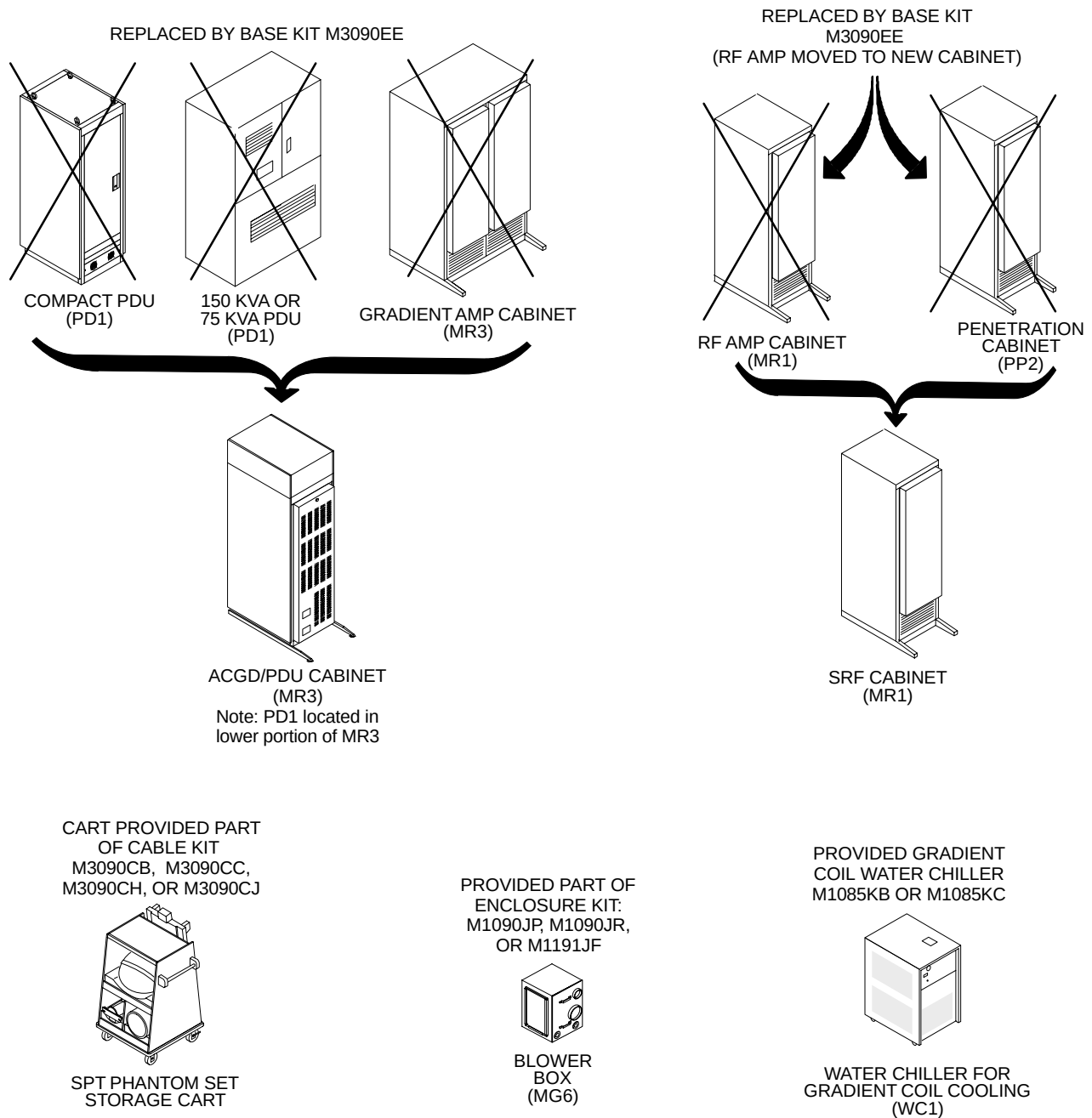


1-2-3 Signa Advantage 1.5T 5.X & 4.X (Continued)



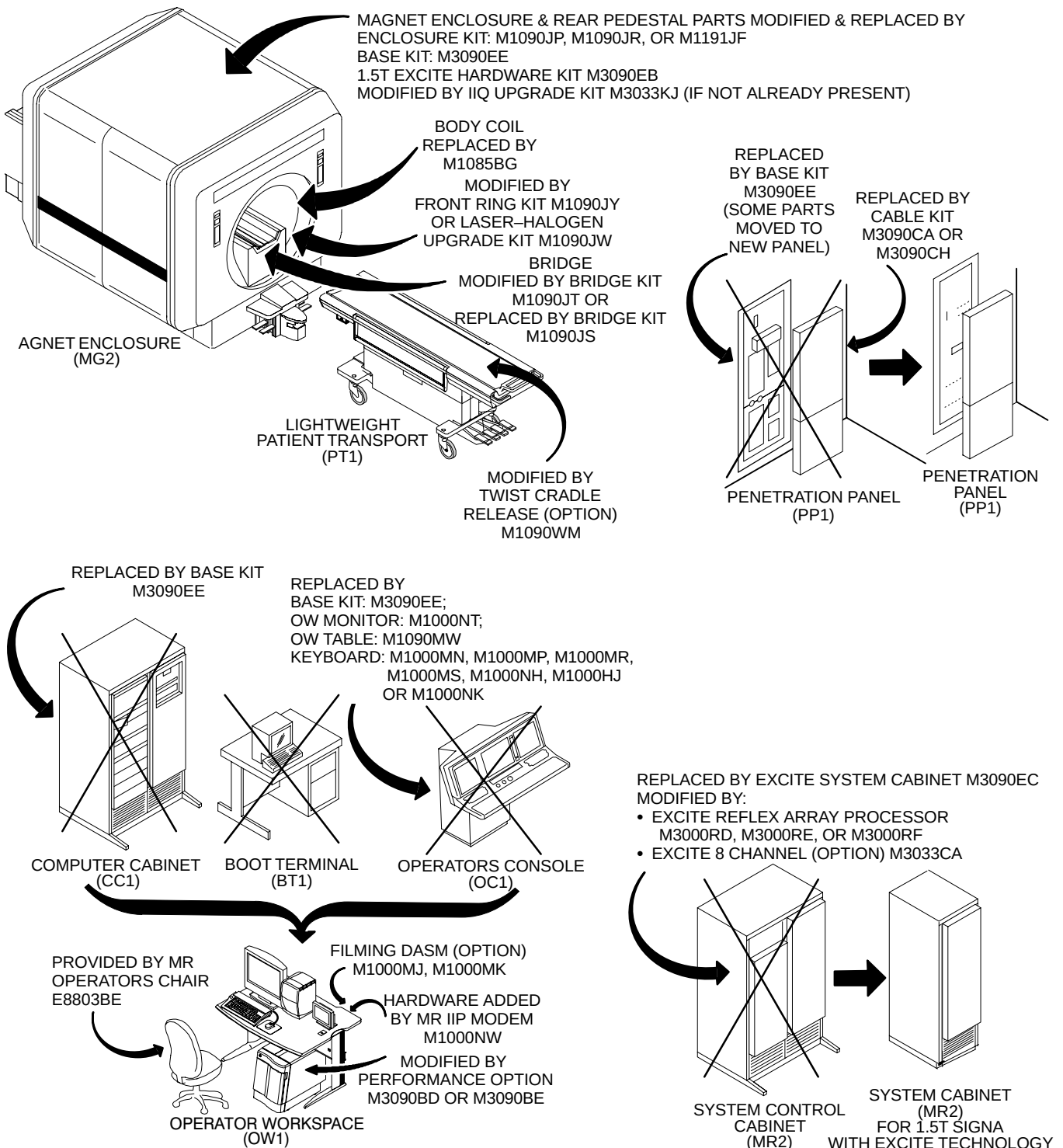
SIGNA HORIZON 5.X SOFTWARE ONLY & SIGNA ADVANTAGE 5.X SYSTEM
EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-13

1-2-3 Signa Advantage 1.5T 5.X & 4.X (Continued)



SIGNA HORIZON 5.X SOFTWARE ONLY & SIGNA ADVANTAGE 5.X SYSTEM
EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
 ILLUSTRATION 1-14

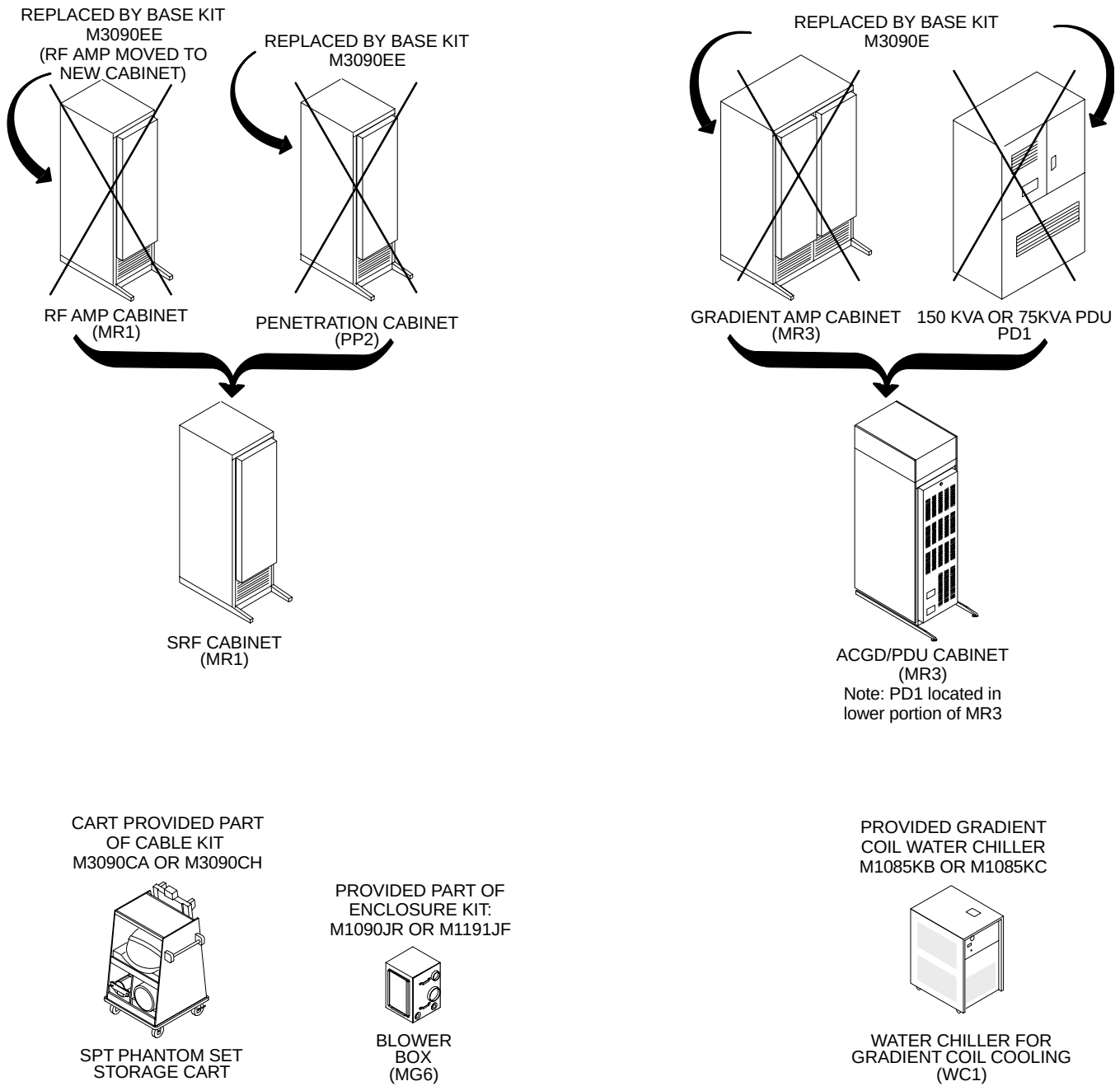
1-2-3 Signa Advantage 1.5T 5.X & 4.X (Continued)



SIGNA ADVANTAGE 4.X SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

ILLUSTRATION 1-15

1-2-3 Signa Advantage 1.5T 5.X & 4.X (Continued)



SIGNA ADVANTAGE 4.X SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-16

1-3 UPGRADES TO SIGNA LX WITH ACGD

Illustration 1-17 shows the system's major equipment after a Signa LX with ACGD upgrade is complete. The upgrade content is dependent on the configuration of the system being upgraded and the final Signa LX with ACGD product configuration. The following sub-sections provide an overview of the various Signa LX with ACGD upgrade hardware.



WARNING!

ALL SYSTEMS UPGRADING TO SYSTEM CONFIGURATION WITH THE ACGD/PDU CABINET NEED TO INVESTIGATE THAT THE FACILITY TRANSFORMER AND FEEDER WIRES ARE CORRECTLY SIZED. ALSO THE MAIN DISCONNECT CONTROL NEEDS TO BE INVESTIGATED FOR CORRECTLY SIZED WIRES AND RATED COMPONENTS TO MEET THE UPGRADE POWER REQUIREMENTS. REFER SECTION 5 – POWER REQUIREMENTS.

SYSTEMS UPGRADING TO SYSTEM CONFIGURATION WITH ACGD/PDU CABINET WITH AN UNINTERRUPTIBLE POWER SUPPLY (UPS) PROVIDING POWER TO THE ENTIRE SYSTEM NEED TO MAKE SURE THE UPS OPERATION PARAMETERS ARE COMPATIBLE WITH THE SIGNA SYSTEM WITH ACGD POWER AND REGULATION DEMANDS. REFER SECTION 5 – POWER REQUIREMENTS.

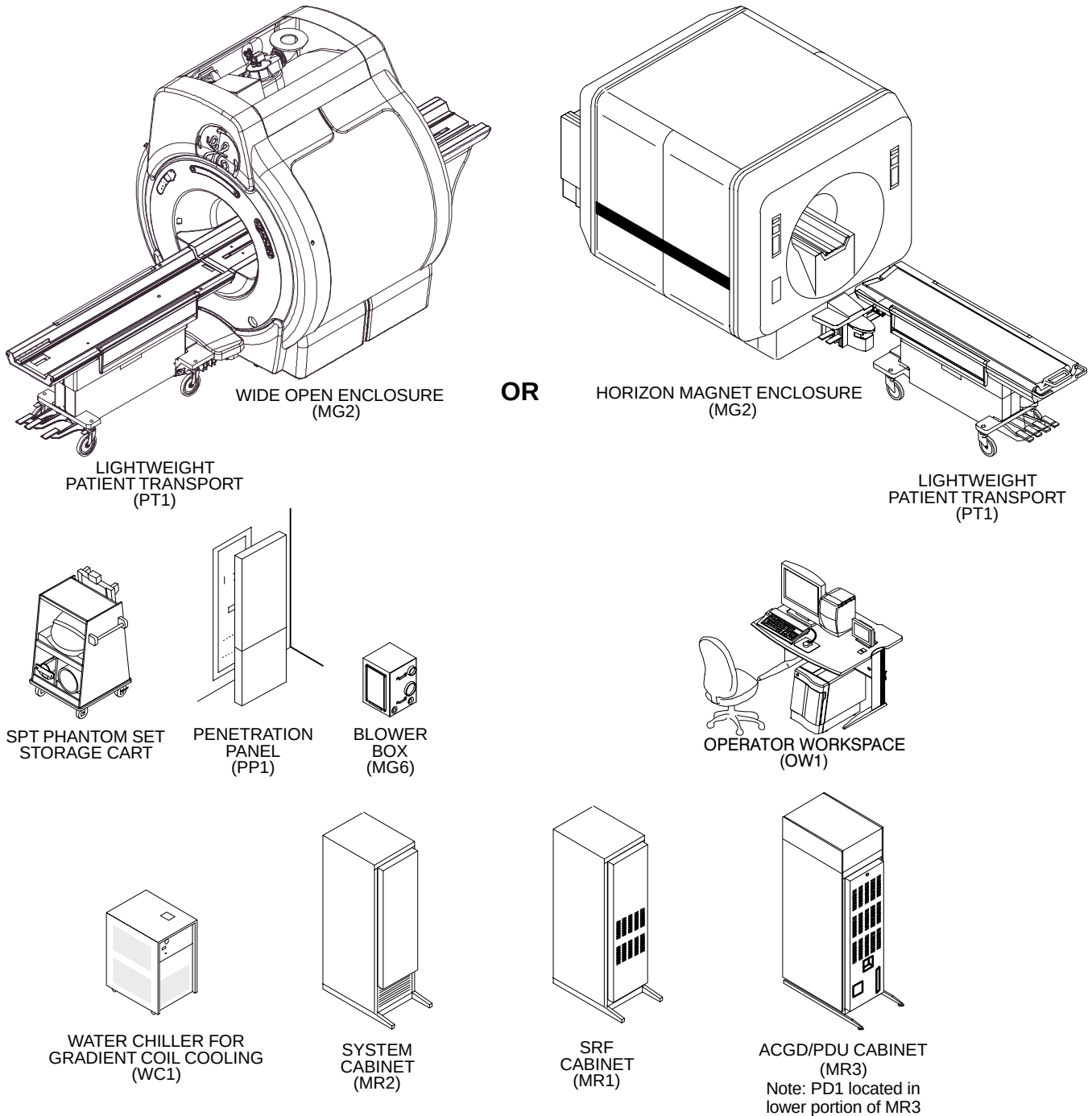
Note

Only catalogs which include hardware are documented in this publication. Software only catalogs are not included, existing system site parameters apply for software only upgrades.

Note

FOR USA, FIXED SITE ONLY CONSIDERING MAGNET UPGRADE: Refer to *Direction 22461012 Pre-Sales/Site Survey For Upgrade To CXK4 Magnet With WideOpen Enclosure* for the many environmental factors and logistics that need to be understood to qualify a site for a magnet upgrade. The Magnet Upgrade Option requires the MR system to be operating a Release 8.X prior or coincident with the Magnet Upgrade option.

1-3 UPGRADES TO SIGNA LX WITH ACGD (Continued)



SIGNA LX WITH ACGD SYSTEM MAJOR EQUIPMENT
ILLUSTRATION 1-17

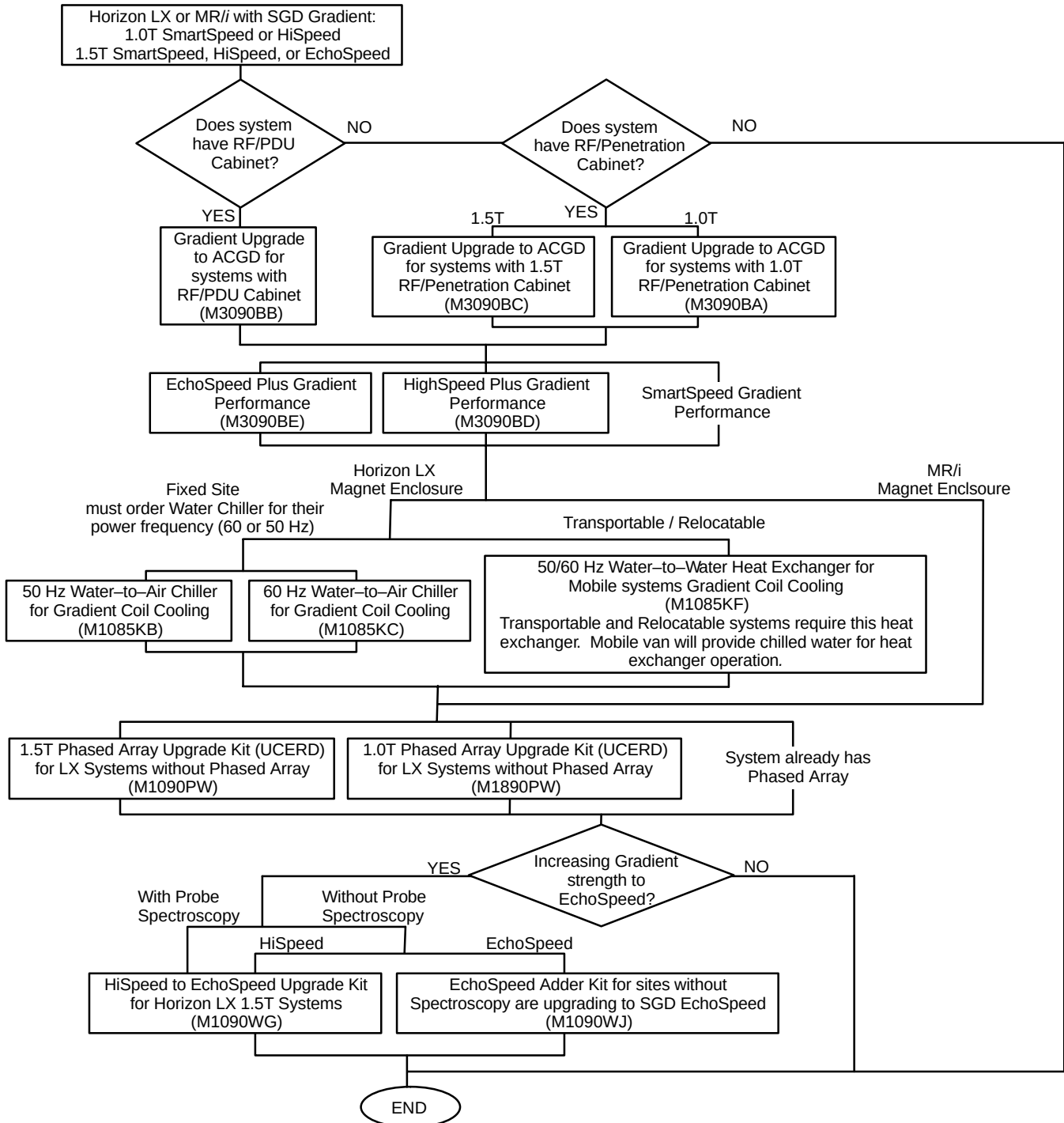
1-3-1 Signa Horizon LX With SGD & Signa MR/i System

Illustration 1-18 lists the upgrade catalogs for Signa Horizon LX with SGD and Signa MR/i with SGD system to Signa LX with ACGD. Illustration 1-19 shows the cabinets replaced and modified by the listed catalogs.

Note

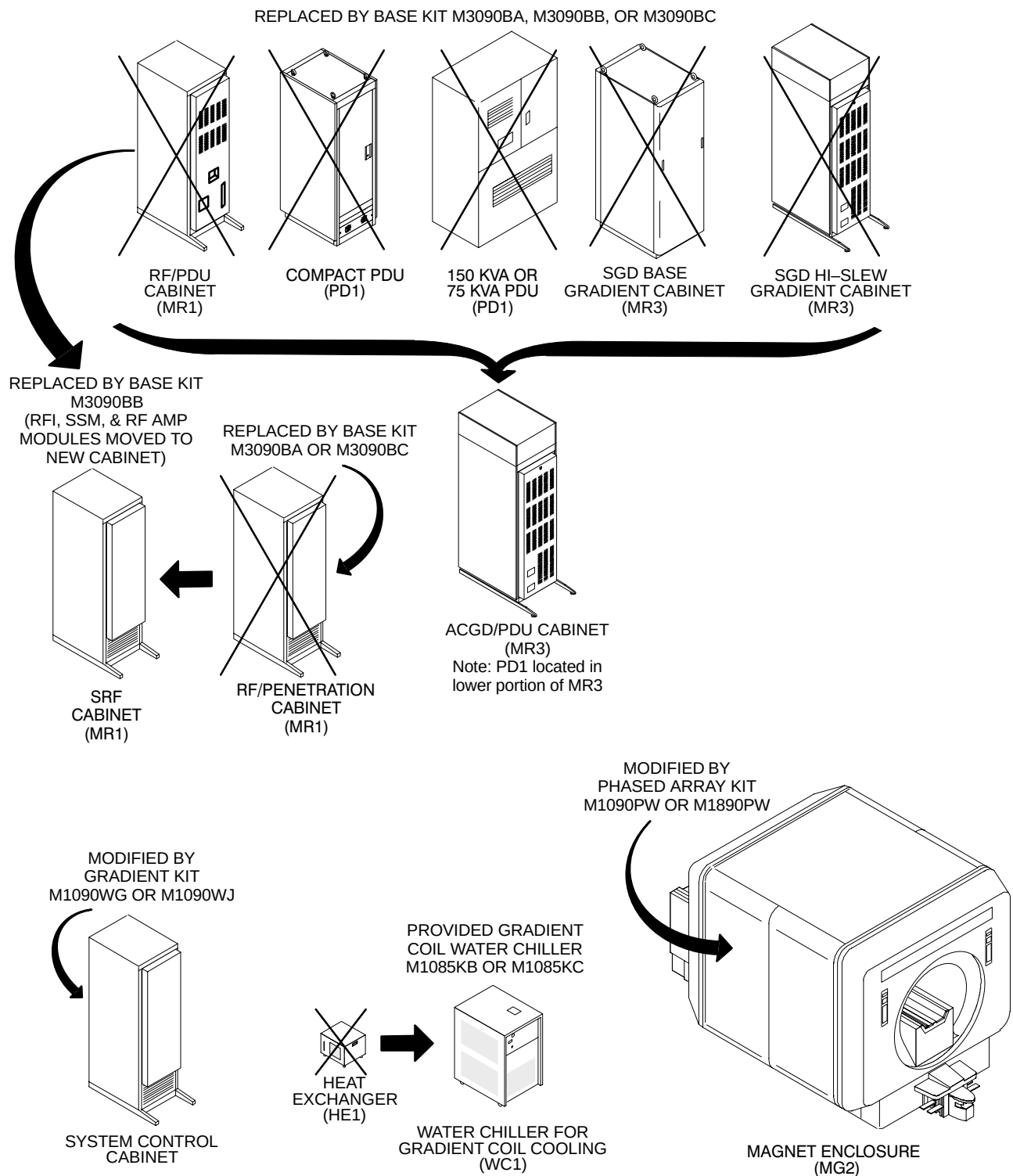
Systems with Oxford or S-1 magnet must upgrade to the LCC magnet before or concurrent with system upgrade.

1-3-1 Signa Horizon LX With SGD & Signa MR/i System (Continued)



1.5T SIGNA LX OR SIGNA MR/i WITH SGD GRADIENTS TO SIGNA LX WITH ACGD CATALOGS

1-3-1 Signa Horizon LX With SGD & Signa MR/i System (Continued)



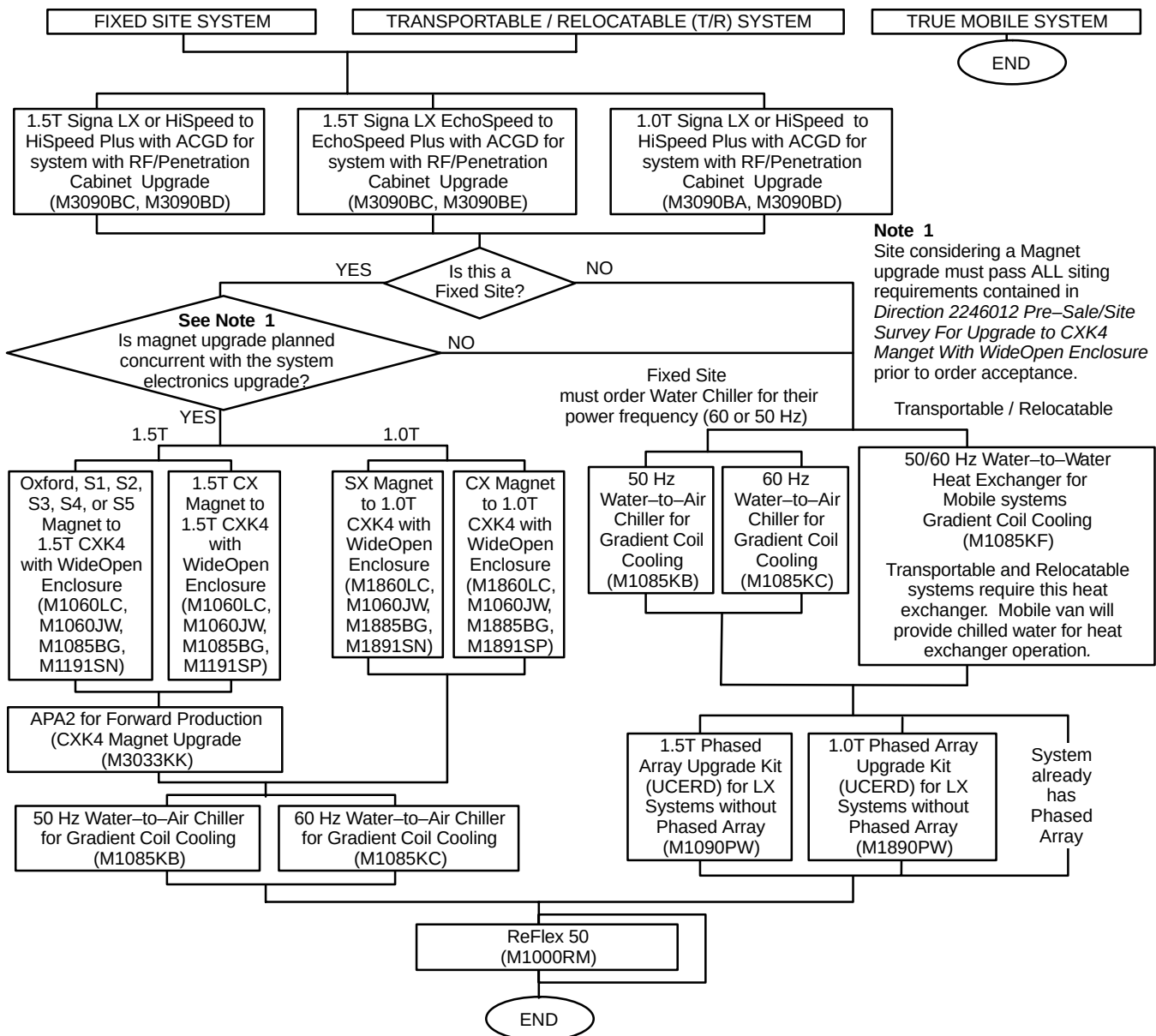
SIGNA HORIZON LX WITH SGD & SIGNA MR// SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-19

1-3-2 Signa Horizon LX With 8645 Gradients System

Illustration 1-20 lists the upgrade catalogs for Signa Horizon LX system with 8645 Gradients to Signa Horizon LX with ACGD. Illustration 1-21 shows the cabinets replaced and modified by the listed catalogs.

Note

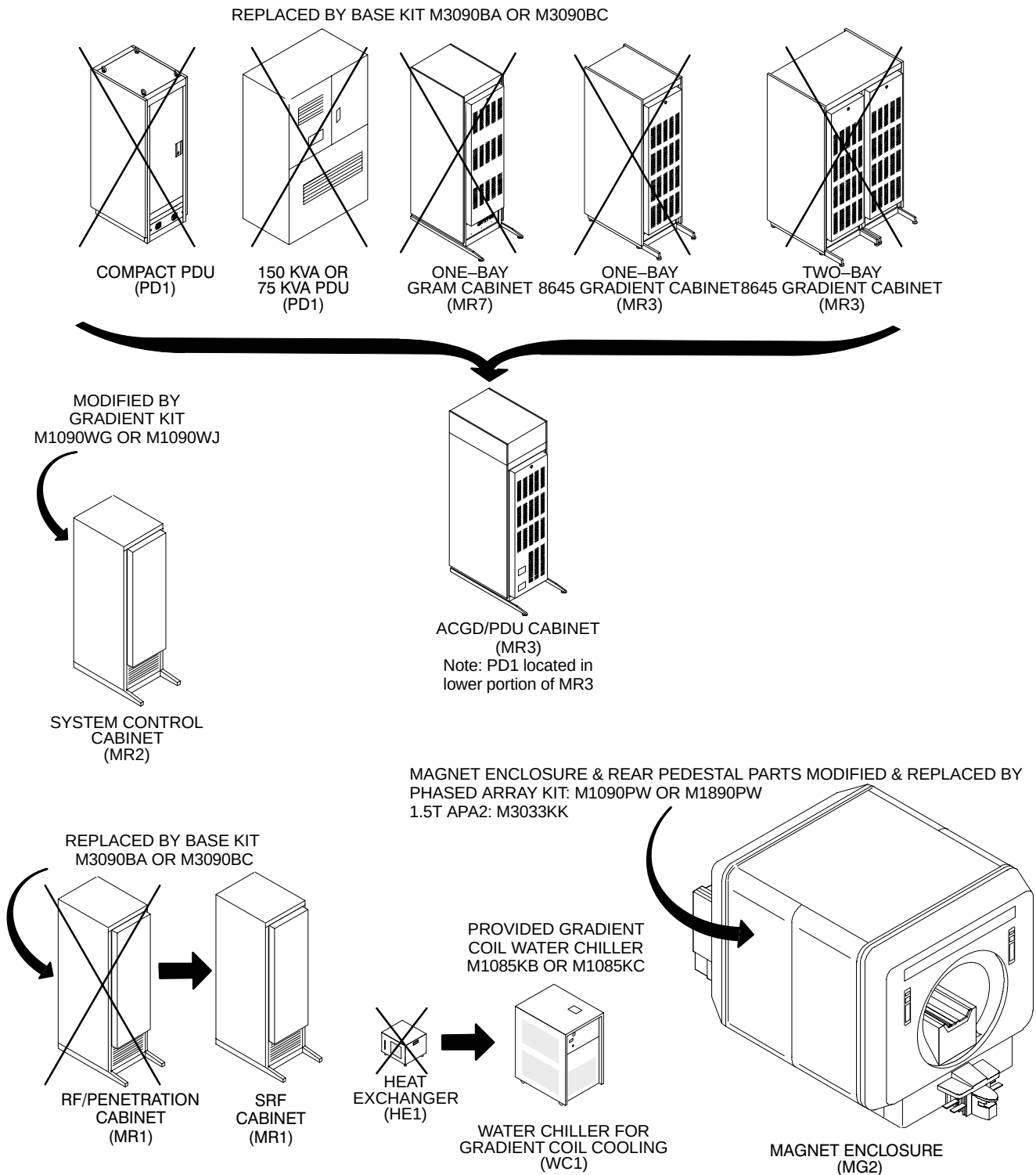
Systems with Oxford or S-1 magnet must upgrade to the LCC magnet before or concurrent with system upgrade.



1.5T SIGNA HORIZON 5.X TO SIGNA LX WITH ACGD CATALOGS

ILLUSTRATION 1-20

1-3-2 Signa Horizon LX With 8645 Gradients System (Continued)



SIGNA HORIZON LX SYSTEM UPGRADE MAJOR EQUIPMENT REPLACED & MODIFIED
ILLUSTRATION 1-21

1-3-3 Signa Horizon 5.X System

■ Illustrations 1-22 through 1-24 list the upgrade catalogs for Signa Horizon 5.X system to Signa Horizon LX with ACGD. Illustrations 1-25 and 1-26 show the cabinets replaced or modified and new equipment added by the listed catalogs.

Refer to Section 1-4 SIGNA HORIZON LX SYSTEM OPTIONS for additional option catalogs.

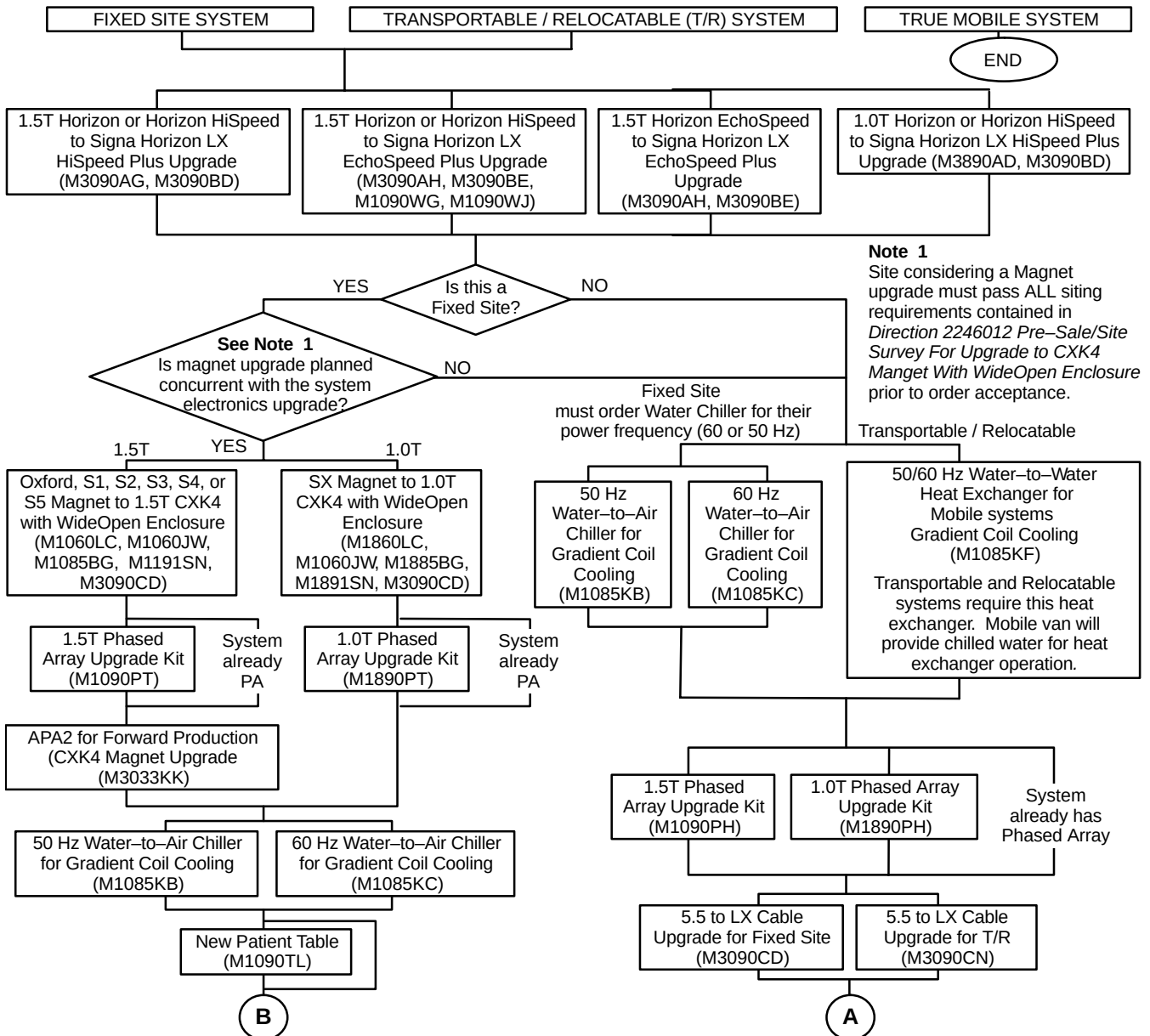
Note

Signa Horizon 5.5 Software only systems have a 55 cm Body Coil and require upgrade catalogs defined in Section 1-3-4, SIGNA HORIZON 5.X SOFTWARE ONLY & SIGNA ADVANTAGE 5.X SYSTEM UPGRADES.

Note

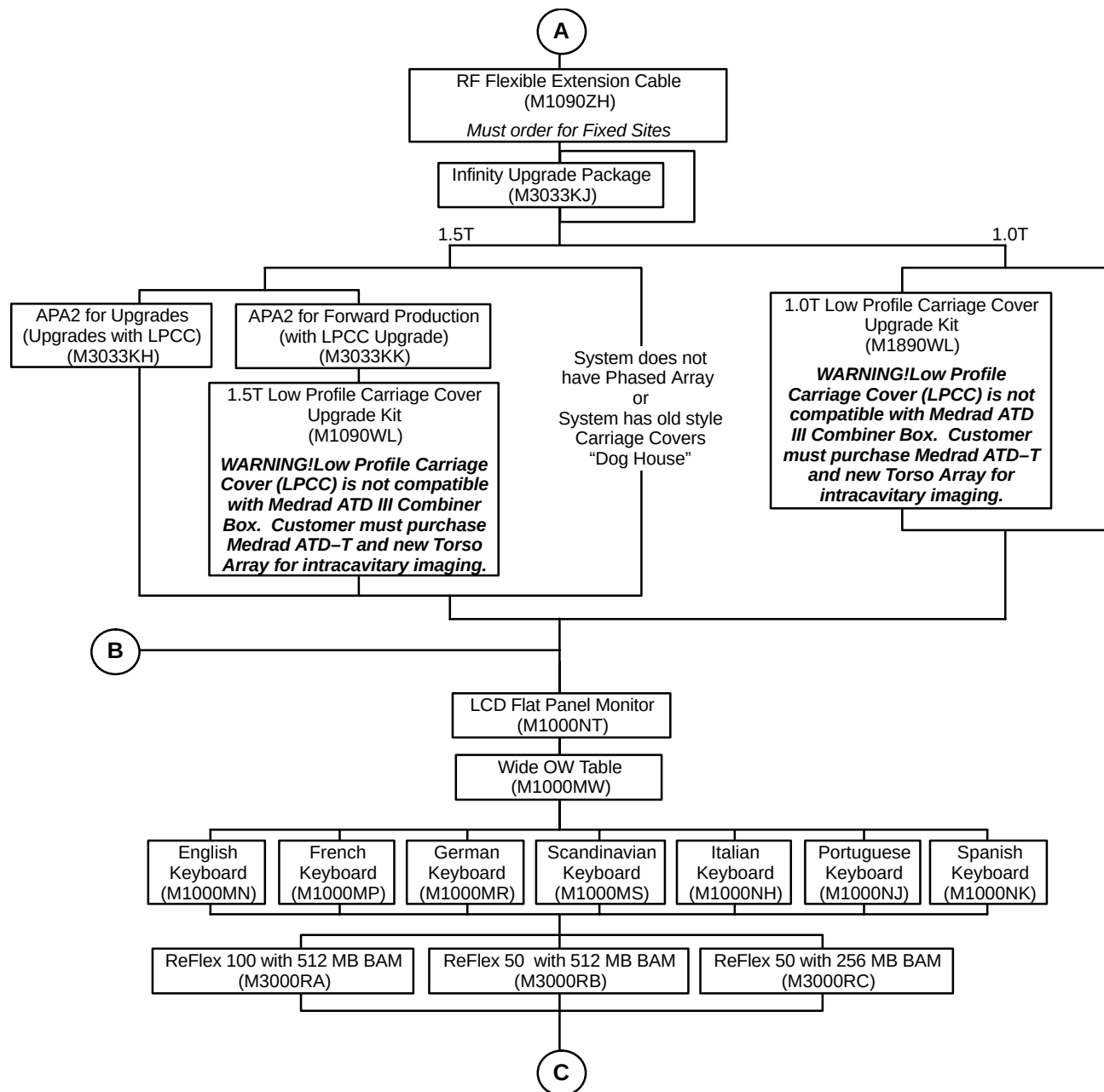
Systems with Oxford or S-1 magnet must upgrade to the LCC magnet before or concurrent with system upgrade.

1-3-3 Signa Horizon 5.X System (Continued)



1.5T SIGNA HORIZON 5.X TO SIGNA LX WITH ACGD CATALOGS
ILLUSTRATION 1-22

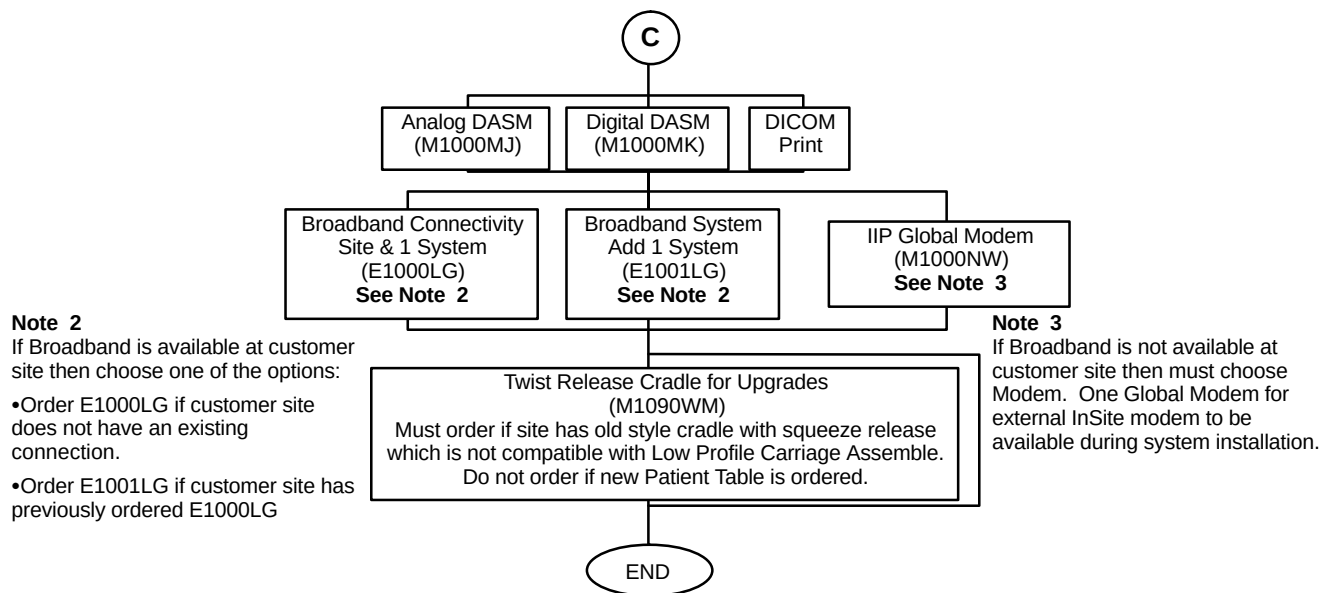
1-3-3 Signa Horizon 5.X System (Continued)



1.5T SIGNA HORIZON 5.X TO SIGNA LX WITH ACGD CATALOGS

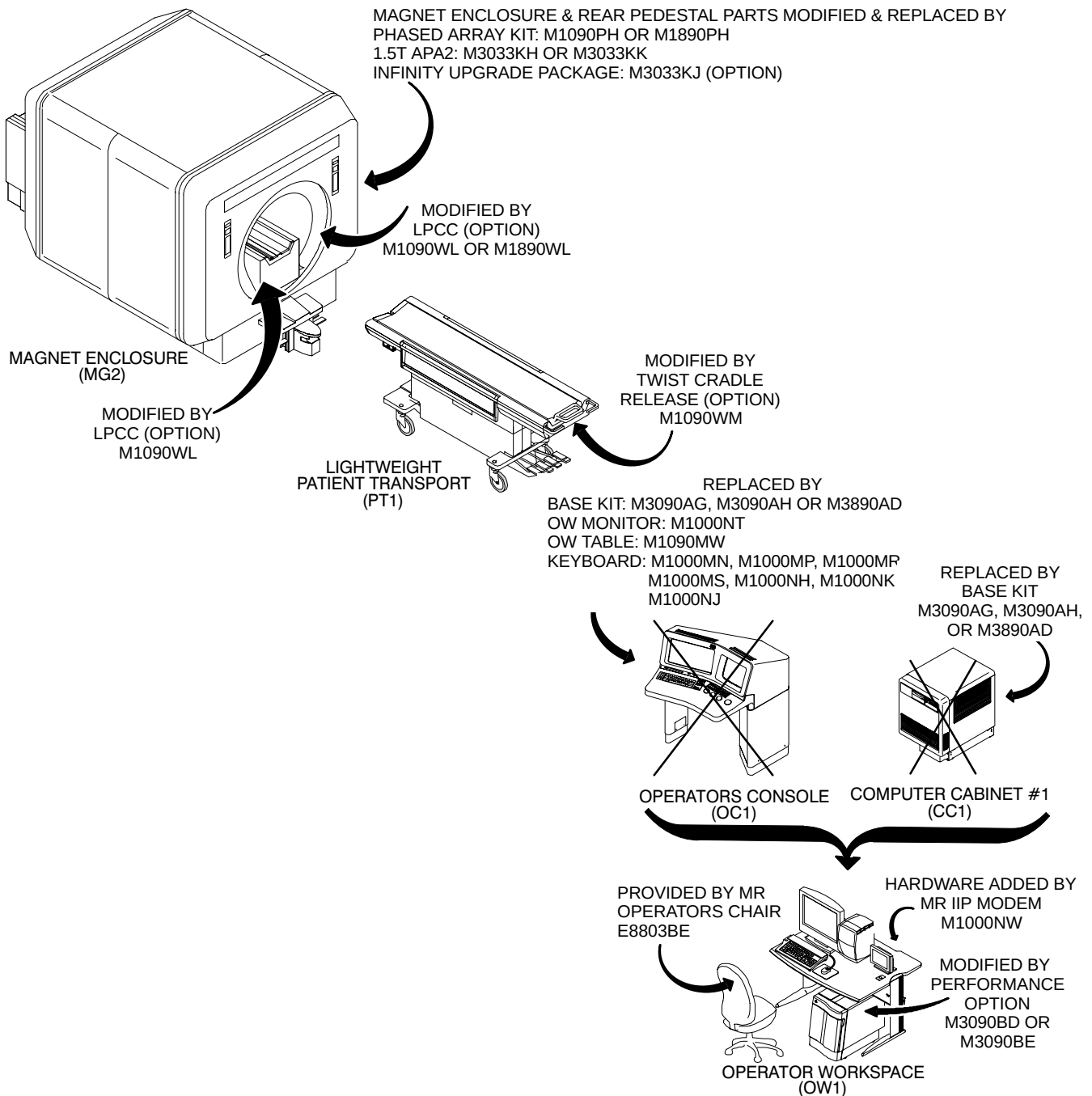
ILLUSTRATION 1-23

■ 1-3-3 Signa Horizon 5.X System (Continued)



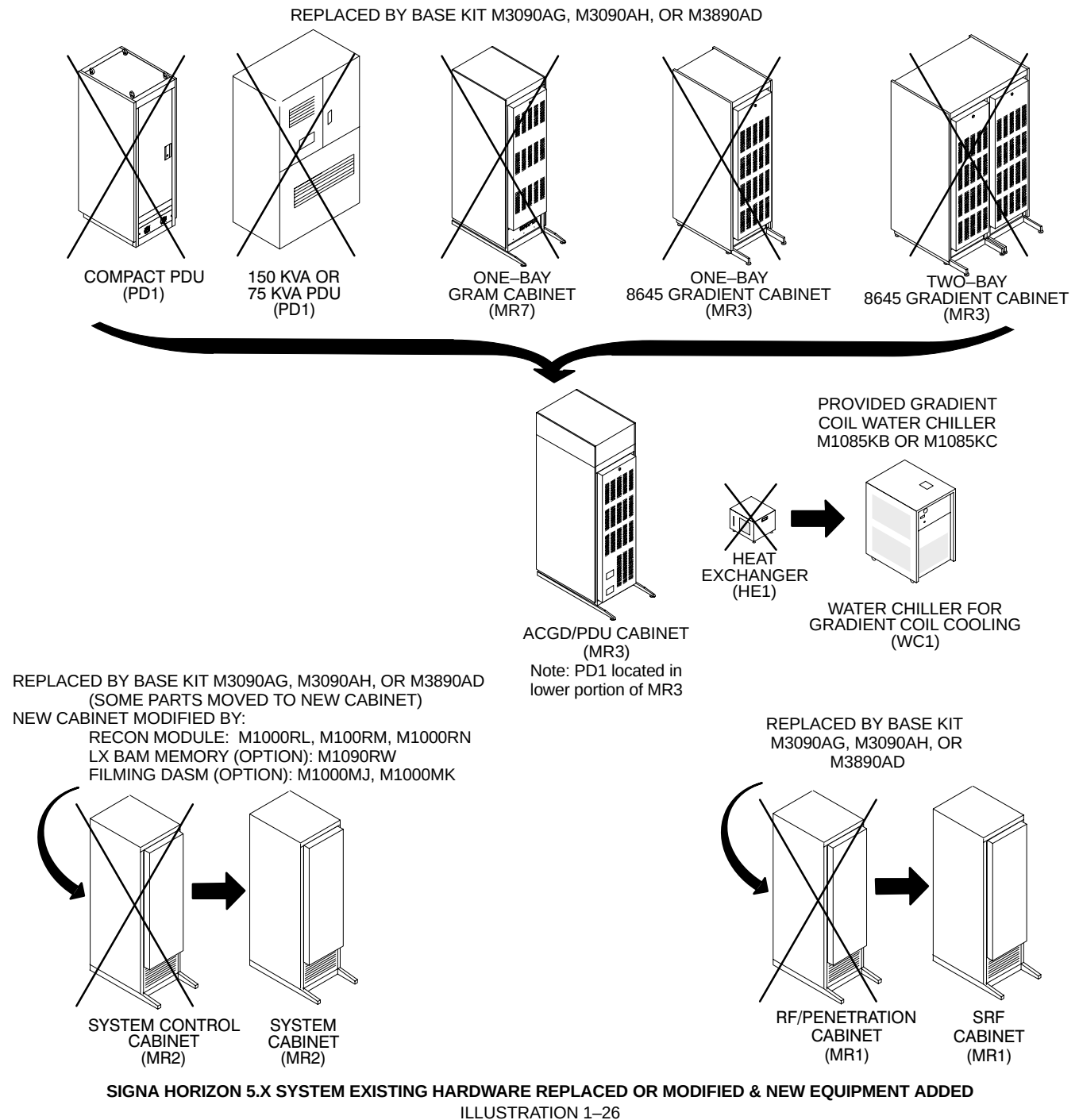
1.5T SIGNA HORIZON 5.X TO SIGNA LX WITH ACGD CATALOGS
ILLUSTRATION 1-24

1-3-3 Signa Horizon 5.X System (Continued)



SIGNA HORIZON 5.X SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-25

1-3-3 Signa Horizon 5.X System (Continued)



1–3–4 Signa Horizon 5.X Software Only & Signa Advantage 5.X System

Illustrations 1–27 through 1–29 lists the upgrade catalogs for Signa Advantage 5.X hardware system and Signa Horizon 5.X software only system to Signa LX with ACGD. Illustrations 1–30 and 1–31 show the cabinets replaced or modified and new equipment added by the listed catalogs. Signa 5.1–5.4 hardware systems are the following configurations:

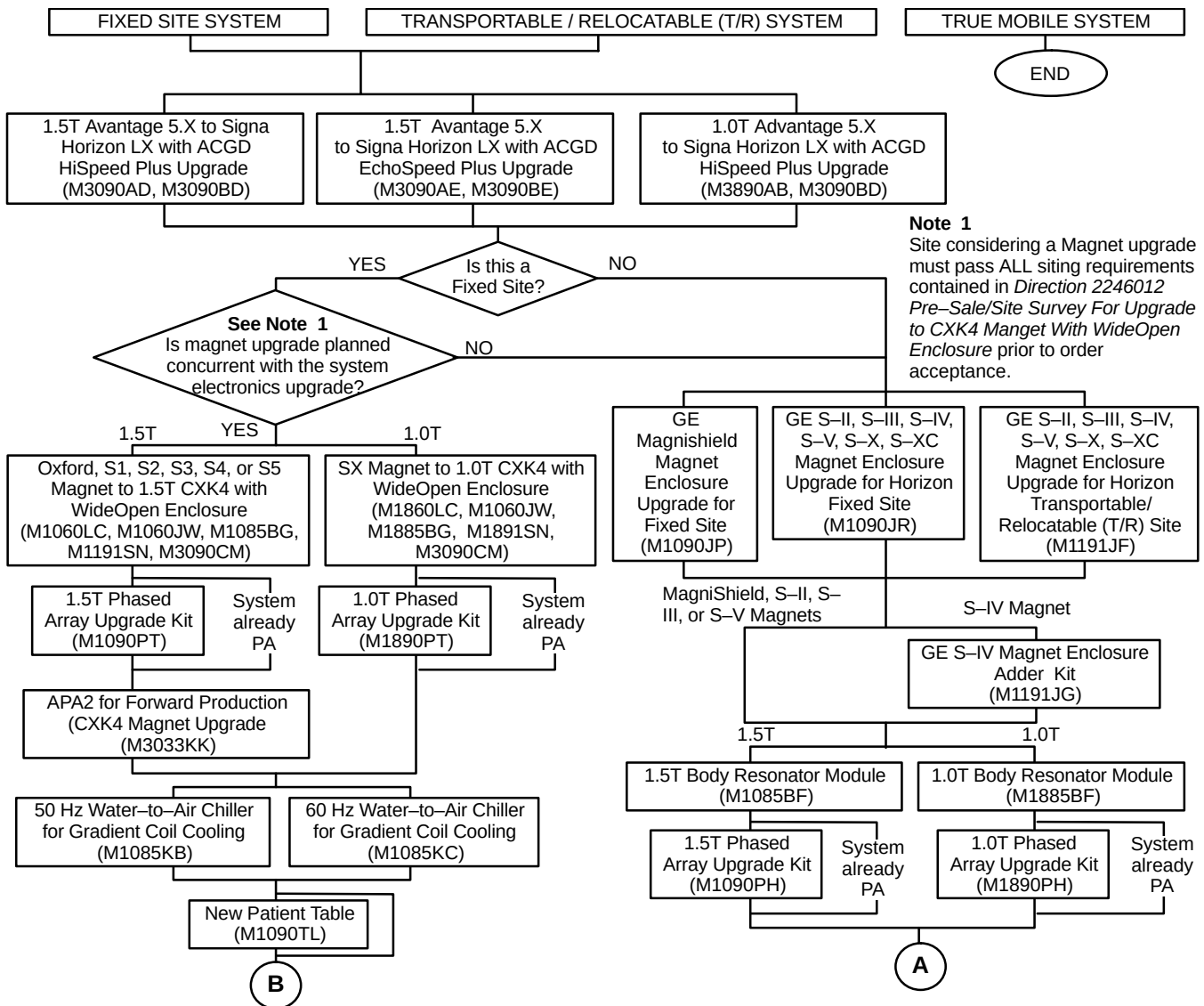
- Signa Horizon 5.X Software only systems which have a 55 cm Body Coil
- Signa Advantage 5.X systems identified by the Genesis Computer, 1-bay System Cabinet, and 8607 Gradient Cabinet.

Refer to Section 1–4 SIGNA HORIZON LX SYSTEM OPTIONS for additional option catalogs.

Note

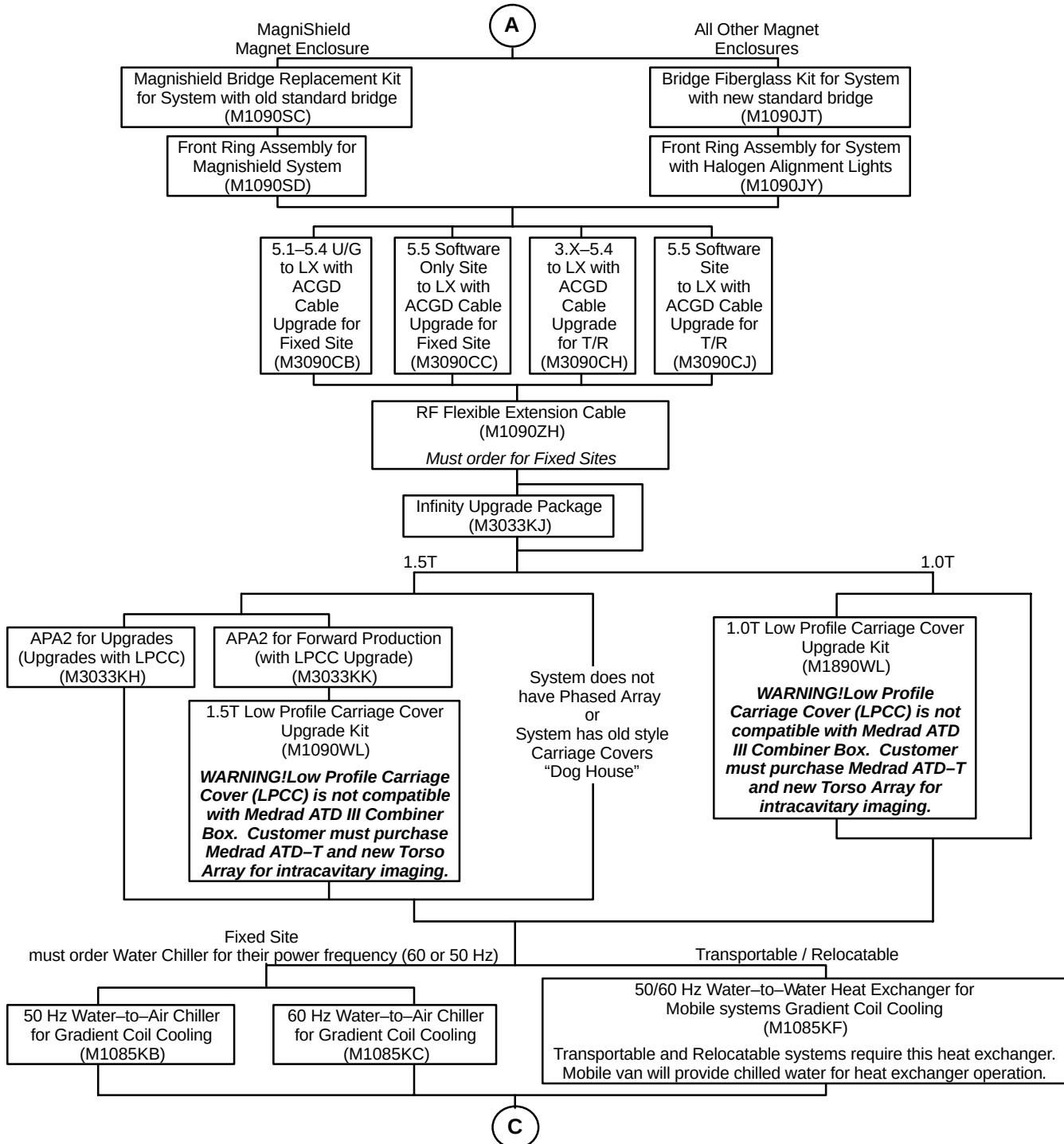
Systems with Oxford or S–1 magnet must upgrade to the LCC magnet before or concurrent with system upgrade.

1-3-4 Signa Horizon 5.X Software Only & Signa Advantage 5.X System – (Continued)



SIGNA HORIZON 5.X SOFTWARE ONLY & SIGNA ADVANTAGE 5.X TO SIGNA LX WITH ACGD CATALOGS
ILLUSTRATION 1-27

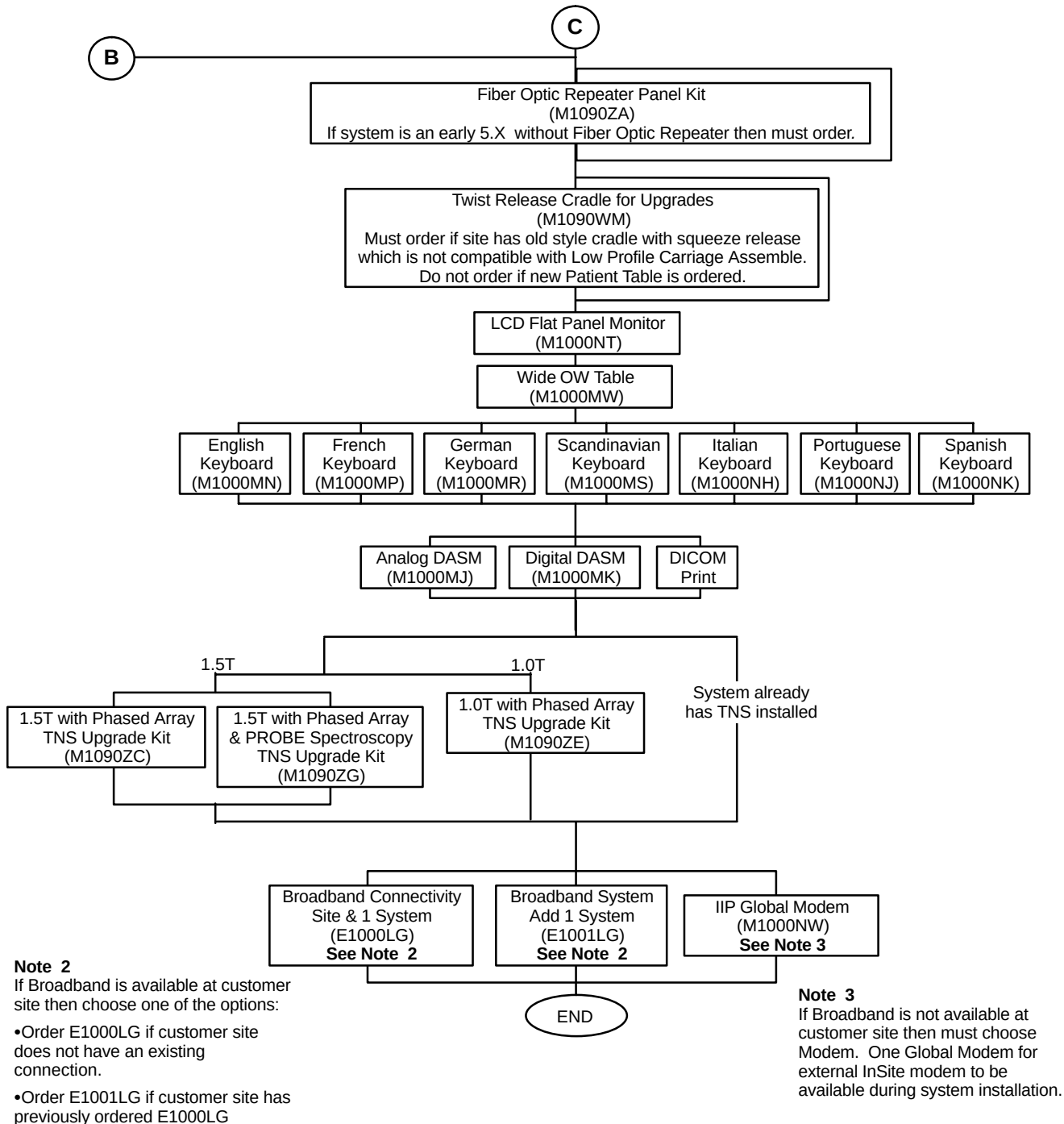
1-3-4 Signa Horizon 5.X Software Only & Signa Advantage 5.X System – (Continued)



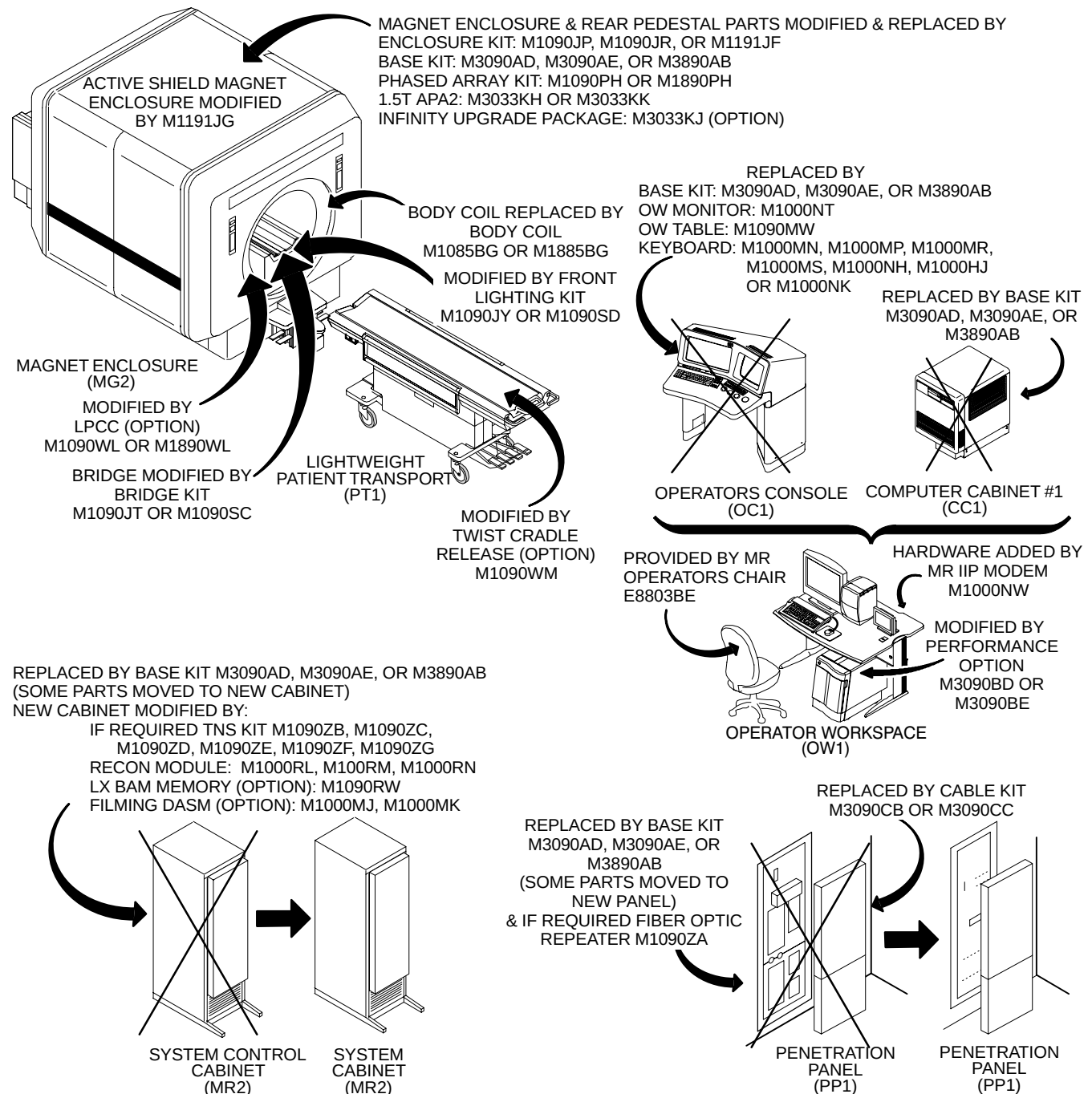
SIGNA HORIZON 5.X SOFTWARE ONLY & SIGNA ADVANTAGE 5.X TO SIGNA LX WITH ACGD CATALOGS

ILLUSTRATION 1–28

1-3-4 Signa Horizon 5.X Software Only & Signa Advantage 5.X System – (Continued)



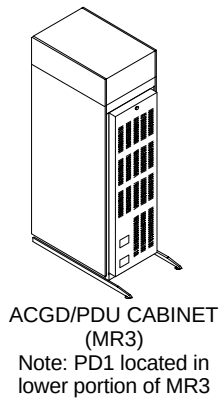
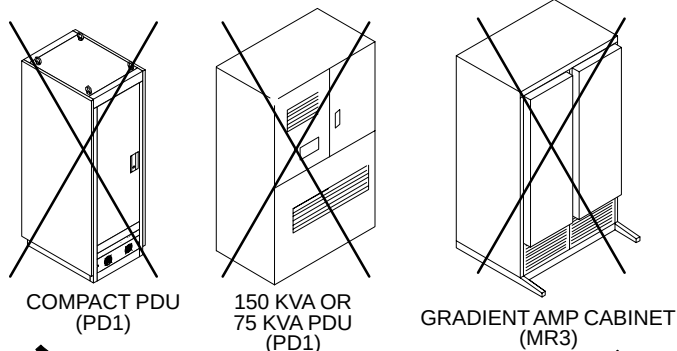
1-3-4 Signa Horizon 5.X Software Only & Signa Advantage 5.X System – (Continued)



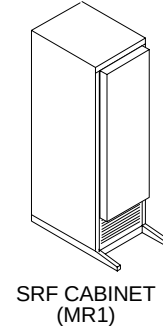
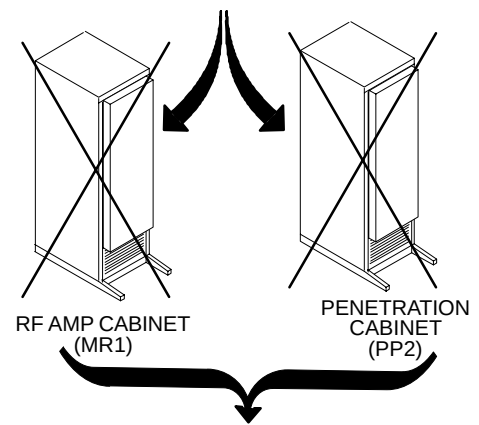
SIGNA HORIZON 5.X SOFTWARE ONLY & SIGNA ADVANTAGE 5.X SYSTEM
EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
 ILLUSTRATION 1-30

1-3-4 Signa Horizon 5.X Software Only & Signa Advantage 5.X System – (Continued)

REPLACED BY BASE KIT M3090AD, M3090AE, OR M3809AB



REPLACED BY BASE KIT M3090AD, M3090AE, OR M3890AB (RF AMP MOVED TO NEW CABINET)



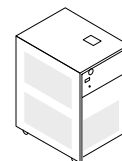
CART PROVIDED PART OF CABLE KIT M3090CB, M3090CC, M3090CH, OR M3090CJ



PROVIDED PART OF ENCLOSURE KIT: M1090JP, M1090JR, OR M1191JF



PROVIDED GRADIENT COIL WATER CHILLER M1085KB OR M1085KC



SIGNA HORIZON 5.X SOFTWARE ONLY & SIGNA ADVANTAGE 5.X SYSTEM
EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
 ILLUSTRATION 1-31

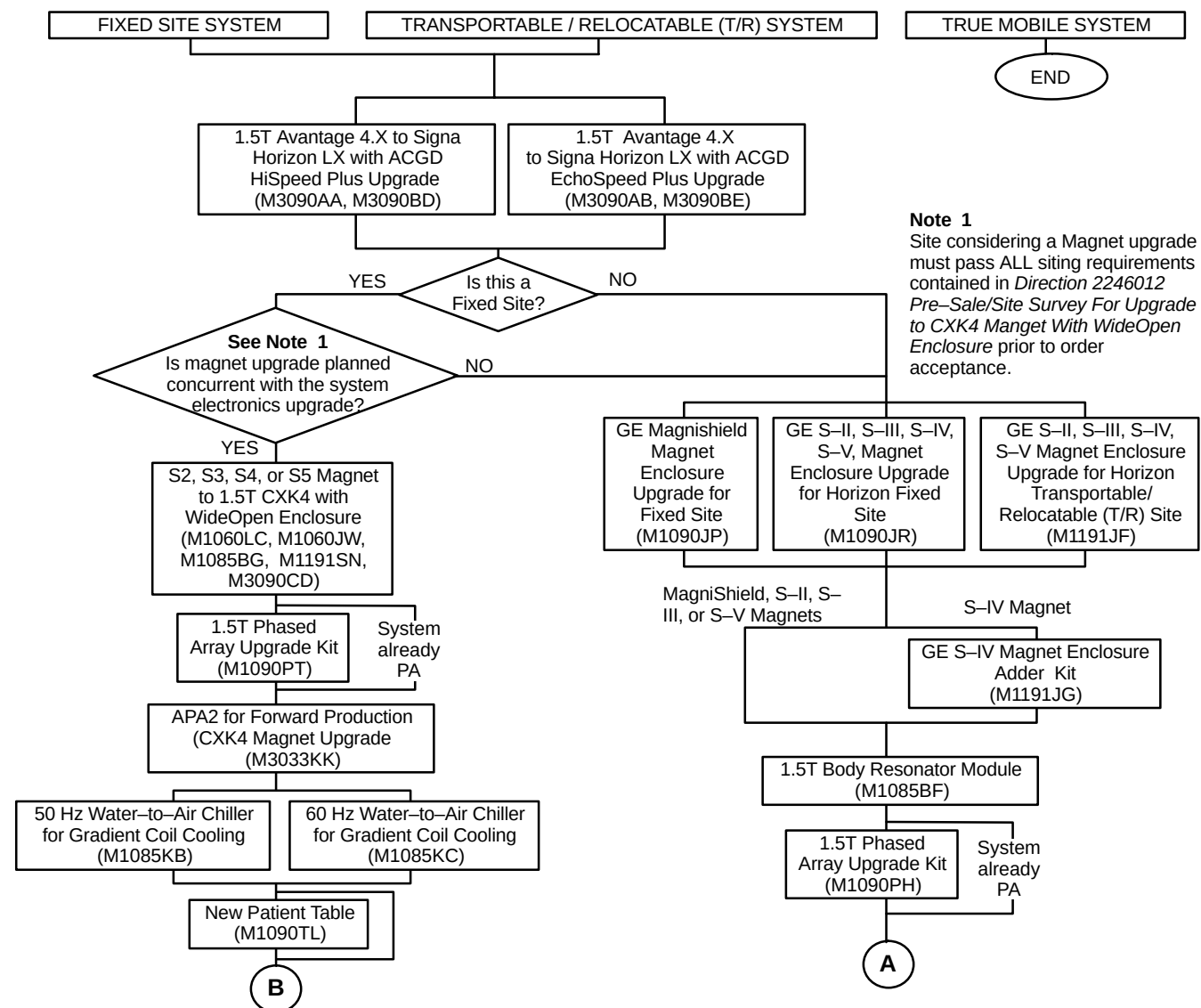
1-3-5 Signa Advantage 4.X System

Illustrations 1-32 through 1-34 lists the upgrade catalogs for Signa Advantage 4.X to Signa Horizon LX with ACGD. Signa Advantage 4.X hardware systems are Signa Advantage 4.5 to 4.8 which can be identified by RF Cabinet model number 46-306900 and Signa Advantage 4.0 to pre-4.5 which can be identified by RF Cabinet model number 46-282900. Illustrations 1-35 and 1-36 show the major equipment replaced or modified and new equipment added by the listed catalogs.

Refer to Section 1-4 SIGNA HORIZON LX SYSTEM OPTIONS for options catalogs.

Note

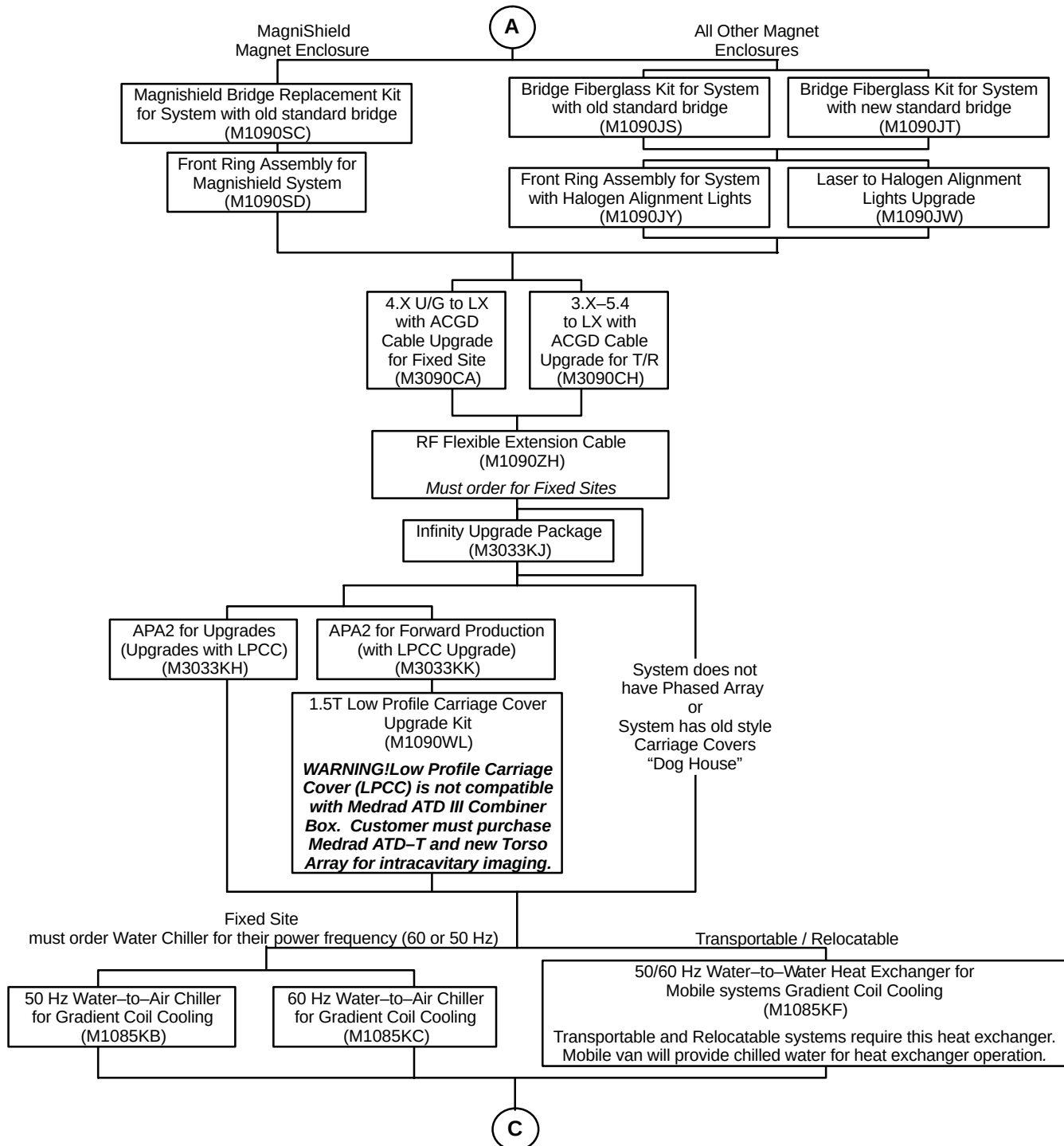
Systems with Oxford or S-1 magnet must upgrade to the LCC magnet before or concurrent with system upgrade.



SIGNA ADVANTAGE 4.X TO SIGNA LX WITH ACGD CATALOGS

ILLUSTRATION 1-32

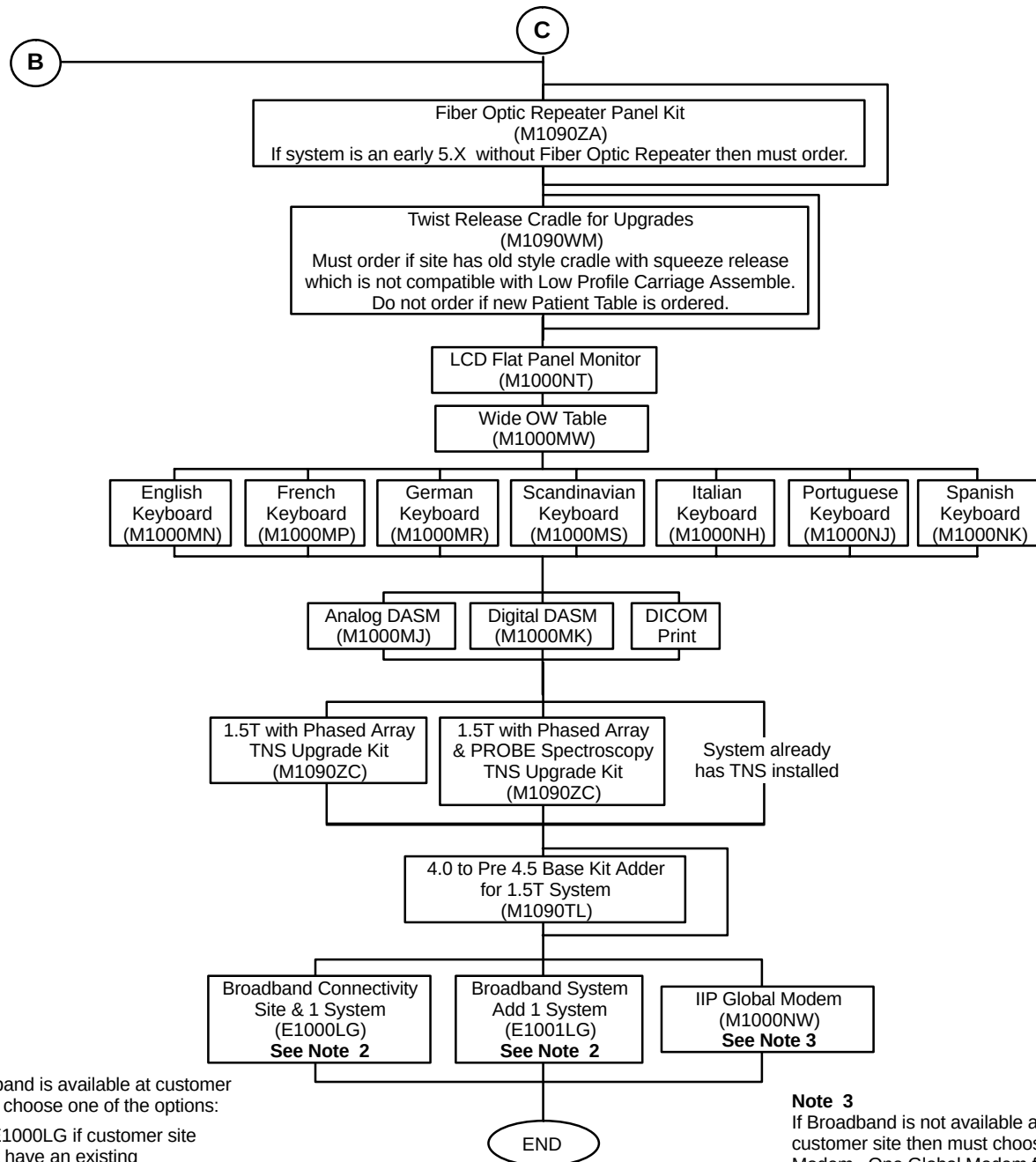
1-3-5 Signa Advantage 4.X System – (Continued)



SIGNA ADVANTAGE 4.X TO SIGNA LX WITH ACGD CATALOGS

ILLUSTRATION 1-33

1-3-5 Signa Advantage 4.X System – (Continued)



Note 2

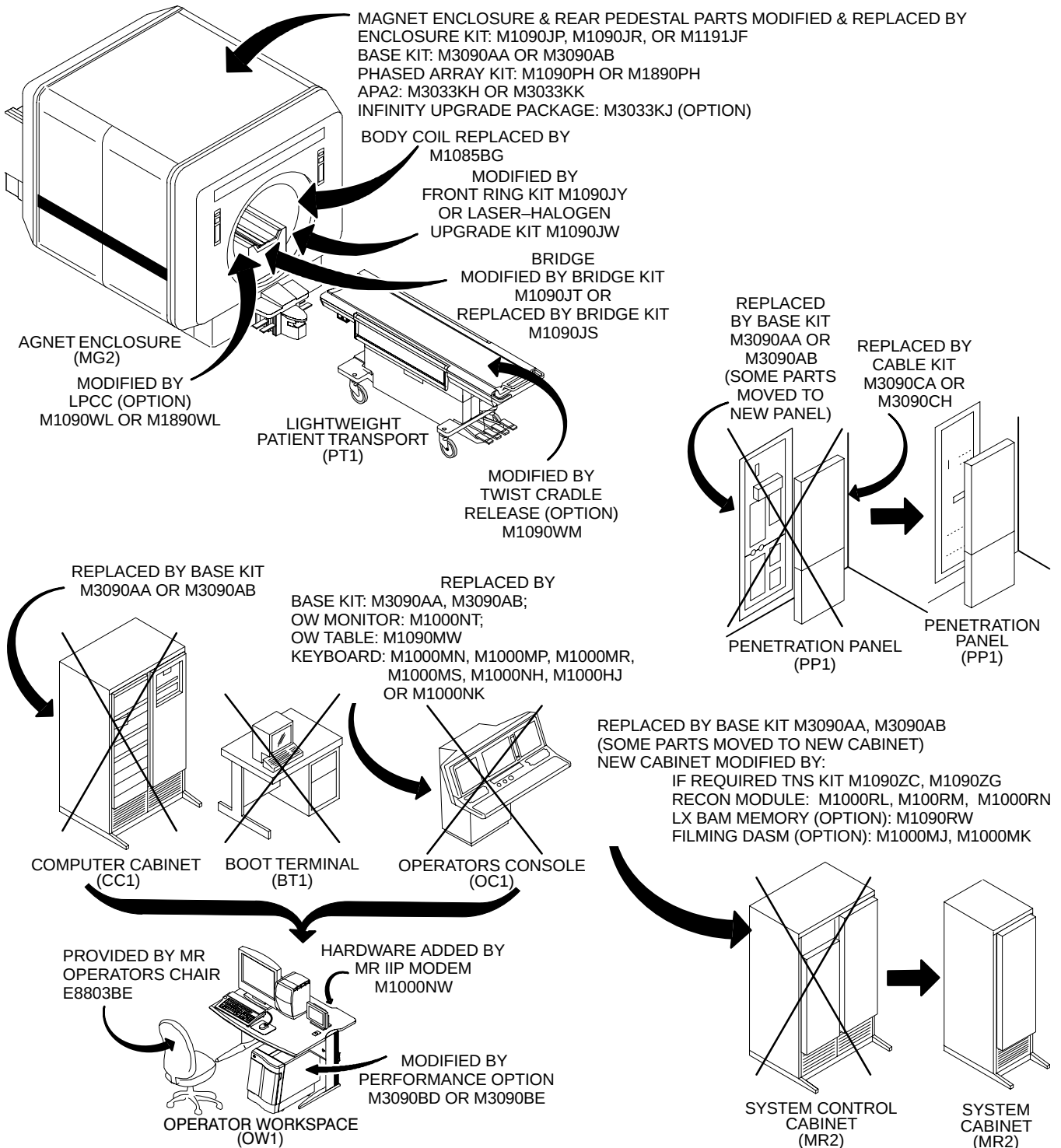
If Broadband is available at customer site then choose one of the options:

- Order E1000LG if customer site does not have an existing connection.
- Order E1001LG if customer site has previously ordered E1000LG

Note 3

If Broadband is not available at customer site then must choose Modem. One Global Modem for external InSite modem to be available during system installation.

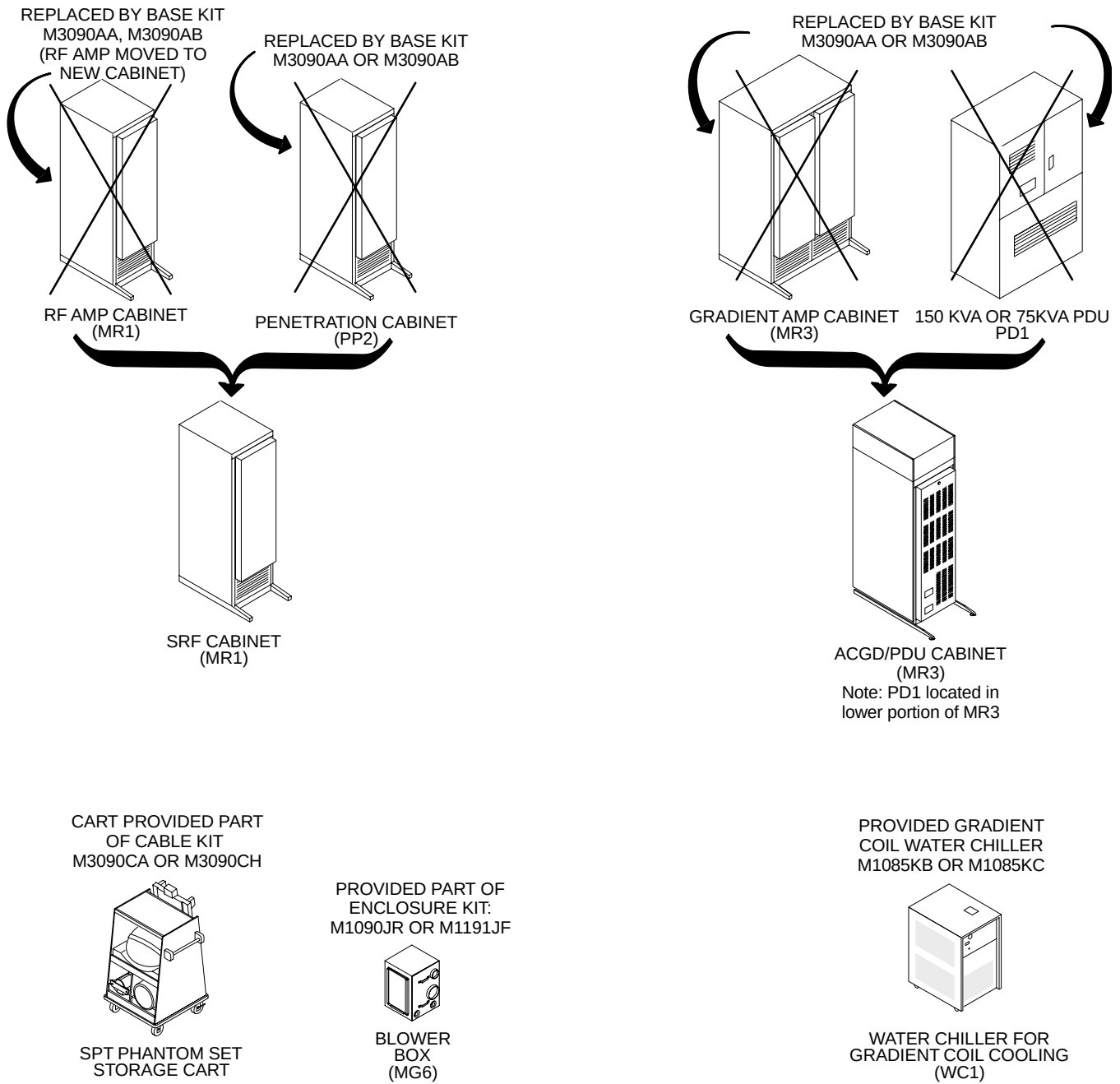
1-3-5 Signa Advantage 4.X System (Continued)



SIGNA ADVANTAGE 4.X SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

ILLUSTRATION 1-3-5

1-3-5 Signa Advantage 4.X System (Continued)



SIGNA ADVANTAGE 4.X SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-36

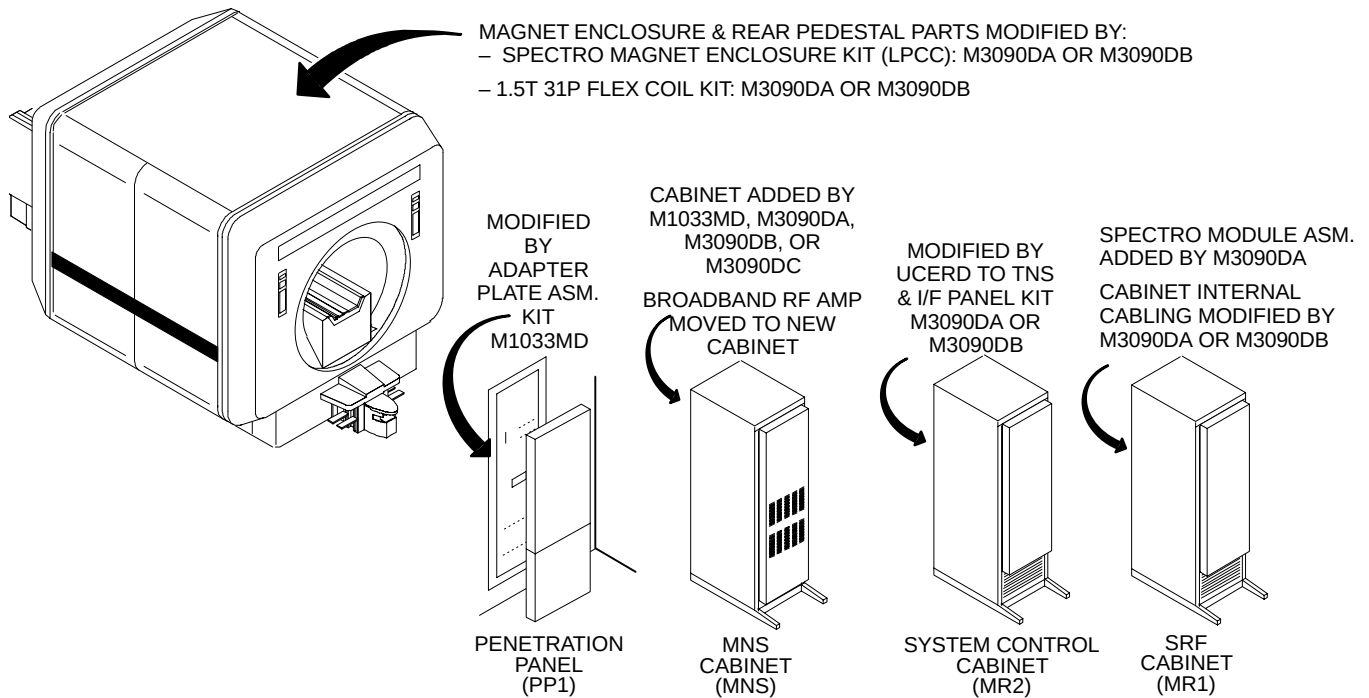
1-4 SIGNA LX SYSTEM OPTIONS

Signa LX system options which have site preparation impact include the following.

- VCR Interface (E8803BS, E8801YA, E8801YD)
- Remote Keyboard Option for use with Operator Workspace PC LCD:
 - English (E6901A)
 - German (E6901AA)
 - Spanish (E6901AB)
 - Italian (E6901AC)
 - French (E6901AD)
 - Scandinavian (E6901AE)
- Bar Code Reader (M1090PM) for use in conjunction with HIS/RIS software option.
- For 1.5T system only: Spectroscopy – Probe2000 (M1033JD) is compatible with all 1.5T Signa LX systems. Probe2000 includes both Single and Multi-Voxel Proton Spectroscopy techniques as well as Metabolic Map Spectral Display capabilities. The Resistive Shim Kit (M1040BH) is recommended for GE S-II or S-III magnets and should be included in initial site plan. The recommendation for the Resistive Shim Kit should be discussed with MR Spectroscopy Marketing which can be reached at (262) 521-6746.

1-4 SIGNA LX SYSTEM OPTIONS (Continued)

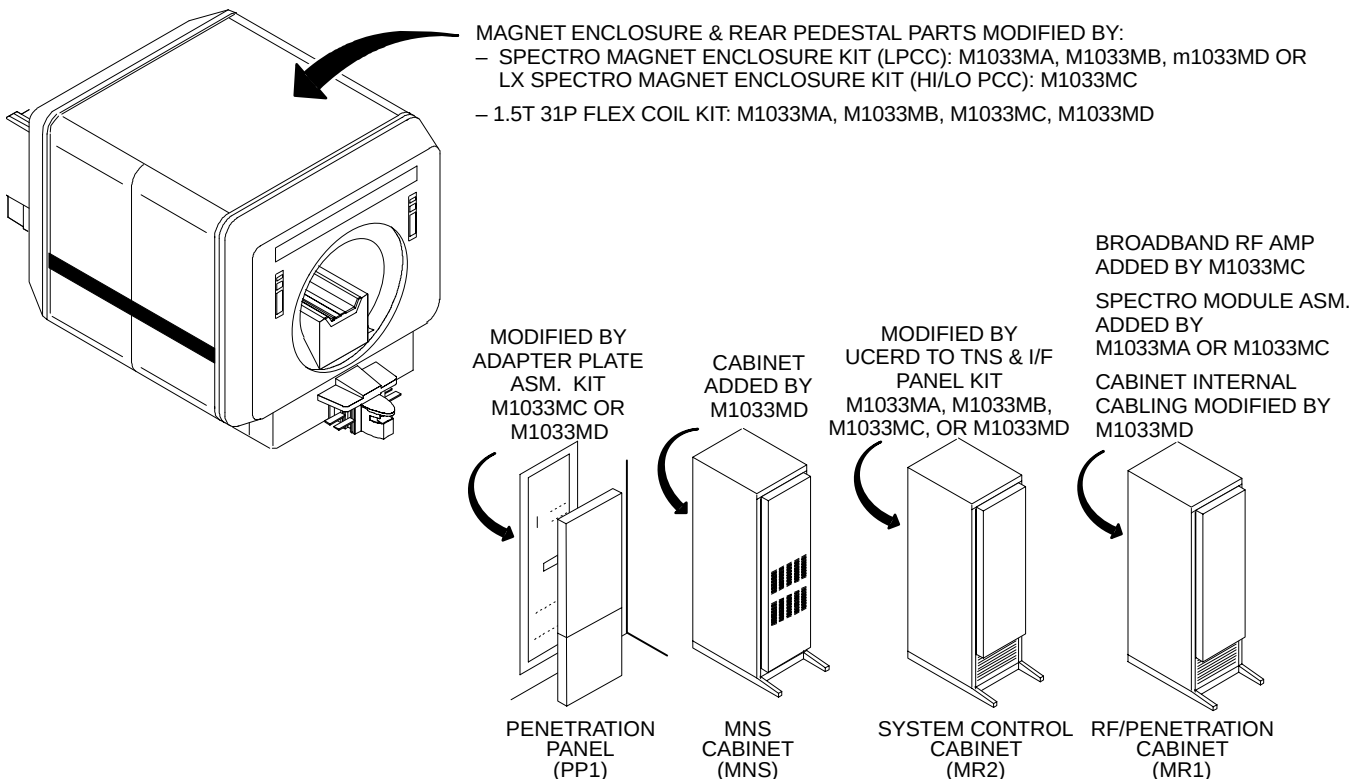
- For 1.5T system ACGD/PDU Gradient Cabinet: The following Multi-Nuclear Spectroscopy (MNS) options are available (refer to Illustration 1-37):
 - If system never had Multi-Nuclear Spectroscopy or did not retain all parts from the MNS Kit then need to install MNS Kit For Signa LX Systems That Have RF/PDU Cabinet or ACGD Cabinet (M1033MD). Kit requires UCERD (can be ordered M1033MF), RF/PDU Cabinet or ACGD Cabinet.
 - For Signa Advantage 4.X or Signa Advantage 5.4 system upgrading to Signa LX:
If system already has Multi-Nuclear Spectroscopy and have all the parts on site then can install 8.X MNS Upgrade For Signa Advantage 4.X-5.4 (M3090DA).
 - For Signa Horizon 5.5 or later system upgrading to Signa LX:
If system already has Multi-Nuclear Spectroscopy and have all the parts on site then can install 8.X MNS Upgrade For Signa Horizon 5.5 And Later (M3090DB).
 - Signa LX with Spectroscopy pre-RF/PDU Cabinet to ACGD Upgrade (M3090DC).



1.5T SYSTEM WITH ACGD/PDU CABINET: MNS OPTIONS EXISTING HARDWARE MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-37

1-4 SIGNA LX SYSTEM OPTIONS (Continued)

- For 1.5T system with SGD Cabinets: The following Multi-Nuclear Spectroscopy (MNS) options are available (refer to Illustration 1-37):
 - For Signa Advantage 4.X or Signa Advantage 5.4 system upgrading to Signa LX:
If system already has Multi-Nuclear Spectroscopy and have all the parts on site then can install 8.X MNS Upgrade For Signa Advantage 4.X–5.4 (M1033MA). Kit requires UCERD (can be ordered M1033MF), non-RF/PDU Cabinet, and non-ACGD Cabinet.
 - For Signa Horizon 5.5 or later system upgrading to Signa LX:
If system already has Multi-Nuclear Spectroscopy and have all the parts on site then can install 8.X MNS Upgrade For Signa Horizon 5.5 And Later (M1033MB). Kit requires UCERD (can be ordered M1033MF), non-RF/PDU Cabinet, and non-ACGD Cabinet.
 - If system never had Multi-Nuclear Spectroscopy or did not retain all parts from the MNS Kit then need to install MNS Kit for Signa LX Systems That Do Not Have RF/PDU Cabinet (M1033MC). Kit requires UCERD (can be ordered M1033MF), non-RF/PDU Cabinet, and non-ACGD Cabinet.
 - If system never had Multi-Nuclear Spectroscopy or did not retain all parts from the MNS Kit then need to install MNS Kit For Signa LX Systems That Have RF/PDU Cabinet or ACGD Cabinet (M1033MD). Kit requires UCERD (can be ordered M1033MF), RF/PDU Cabinet or ACGD Cabinet.



1.5T SYSTEM WITH SGD CABINET: MNS OPTIONS EXISTING HARDWARE MODIFIED & NEW EQUIPMENT ADDED

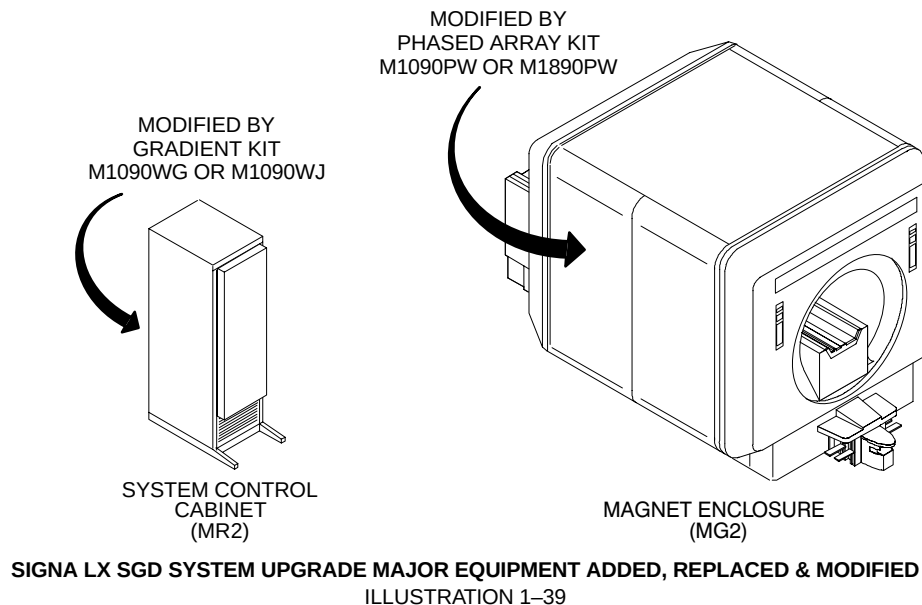
ILLUSTRATION 1-38

1-5 UPGRADES TO SIGNA LX WITH SGD SYSTEM

Table 1-1 lists the upgrade catalogs for Base Signa LX SGD or Signa LX SGD HiSpeed system within the Signa Horizon LX with SGD product line. Illustration 1-21 shows the cabinets added, replaced, and modified by the listed catalogs.

TABLE 1-1
SIGNA LX SGD SYSTEM WITH SGD UPGRADE CATALOGS

SYSTEM HARDWARE CONFIG	UPGRADE KIT	LX WITH SGD UPGRADES CATALOGS	
		HISPEED	ECHOSPEED
Base Horizon LX without Phased Array	Phased Array Kit	M1090PW – 1.5T Phased Array Upgrade Kit (UCERD) for LX Systems without Phased Array M1890PW – 1.0T Phased Array Upgrade Kit (UCERD) for LX Systems without Phased Array	M1090PW – 1.5T Phased Array Upgrade Kit (UCERD) for LX Systems without Phased Array
Horizon LX with SGD HiSpeed with Spectroscopy	Gradient Kit <i>Only required if increasing gradient strength</i>		M1090WG – HiSpeed SGD to EchoSpeed SGD Upgrade Kit for Horizon LX with SGD 1.5T Systems
Horizon LX with SGD without Spectroscopy	Gradient Kit <i>Only required if increasing gradient strength</i>		M1090WG – HiSpeed SGD to EchoSpeed SGD Upgrade Kit for Horizon LX with SGD 1.5T Systems M1090WJ – EchoSpeed Adder Kit for sites without Spectroscopy are upgrading to SGD EchoSpeed



1-6 SIGNA HORIZON 5.5 SYSTEM OPTIONS

- Phase Array Compatibility Kit for 1.5T system (M1090PC) or 1.0T system (M1890PA). The site impact of these kits is the system interconnects, for details refer to Section 6-2-7 M1090PC – 5.5 Phase Array Compatibility Kits For 1.5T System or Section 6-2-17 M1890PA – 5.5 Phased Array Compatibility Kits For 1.0T System.

1-7 TRANSPORTABLE/RELOCATABLE UPGRADE

Transportable and Relocatable MR system Horizon upgrade catalogs are listed in the previous upgrade sections. Note, the cable kits for Transportable/Relocatable include a complete set of system interconnect excluding magnet and option related cables.

In addition to the system, the van must also be upgraded to accommodate the Signa Horizon upgrades. The van/system upgrades can be preformed at the Van Manufacturer facility (preferred location) or at customer site. It is customer responsibility to arrange with the appropriate Van Manufacturer for any upgrade to be performed at site.

- Customer must contact Van Manufacturer to arrange for van/system upgrade at Van Manufacturer facility or customer site. Refer to Section 1-9, VAN MANUFACTURERS CONTACT INFORMATION.
- Customer is responsible for van upgrade costs with Van Manufacturer.
- Customer responsible for van transportation to and from Van Manufacturer facility for upgrades perform there.
- MR system check and calibration performance:
 - For van/system upgrades performed at Van Manufacturer facility, the installation will be checked by MR Manufacturing. Final calibrations will be performed by GE Field Service.
 - For van/system upgrades performed at customer site, the installation will be check and final calibrations will be performed by GE Field Service.

Note

Transportable or Relocatable with Oxford magnet can not be upgraded to Horizon configurations. The Oxford magnet can not be moved with the new body coil installed, serious magnet structural damage will result.

1-8 TRUE MOBILE UPGRADE**Note**

Signa Horizon configurations are not available for 1.5T True Mobile systems with Oxford Magnets due to structural limitations of the Oxford magnet in a mobile environment.

MR system Horizon upgrade catalogs for 1.0T True Mobiles are listed in the previous upgrade sections. Note, the cable kits for True Mobile include a complete set of system interconnect excluding magnet and option related cables.

In addition to the system, the van must also be upgraded to accommodate the Signa Horizon upgrades. The True Mobile van/system upgrade are preformed at the Van Manufacturer facility.

- Customer must contact Van Manufacturer to arrange for van/system upgrade at Van Manufacturer facility. Refer to Section 1-9, VAN MANUFACTURERS CONTACT INFORMATION.
- Customer is responsible for van upgrade costs with Van Manufacturer.
- Customer responsible for van transportation to and from Van Manufacturer facility.
- System check and complete calibrations are performed by MR Manufacturing.

1-9 VAN MANUFACTURERS CONTACT INFORMATION

- Ellis & Watts
4400 Glen Willow Lake Lane
Batavia, OH 45103
Telephone: 513-752-9000
FAX: 513-752-4983

Note

Customers with a MICA Relocatable system need to contact Ellis & Watts.

- PDC Facilities
700 Walnut Ridge Drive, PO Box 900
Hartland, WI 53029-0900
Telephone: 262-367-7700
FAX: 262-367-7744
- SMIT Mobile Equipment
Buys Ballotstraat 6
3261 La Oud-Beijerland
Holland
Telephone: (31) 186-6-14322
FAX: (31) 186-6-19367
- AK Specialty Vehicles
16745 South Lathrop Avenue
Harvey, IL 60426
USA
Telephone: 708-596-5066
FAX: 708-596-2480
AK Specialty Vehicles is also the contact for used Calumet Coach trailers.